

DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Xingzhen Chen, Zhuoping Yang, Jinming Zhuang, Shixin Ji, Sarah Schultz,
Zheng Dong*, Weisong Shi[†] and Peipei Zhou

Brown University; Wayne State University*; University of Delaware[†]

xingzhen_chen@brown.edu
peipei_zhou@brown.edu

<https://peipeizhou-eecs.github.io/>

<https://github.com/arc-research-lab/DORA.git>



U.S. DEPARTMENT
of ENERGY

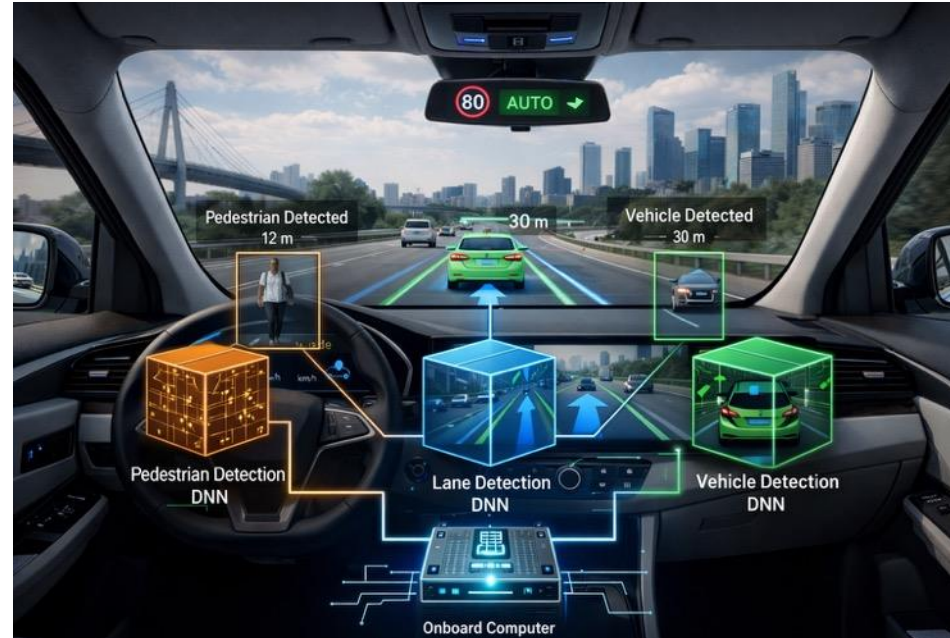


BROWN



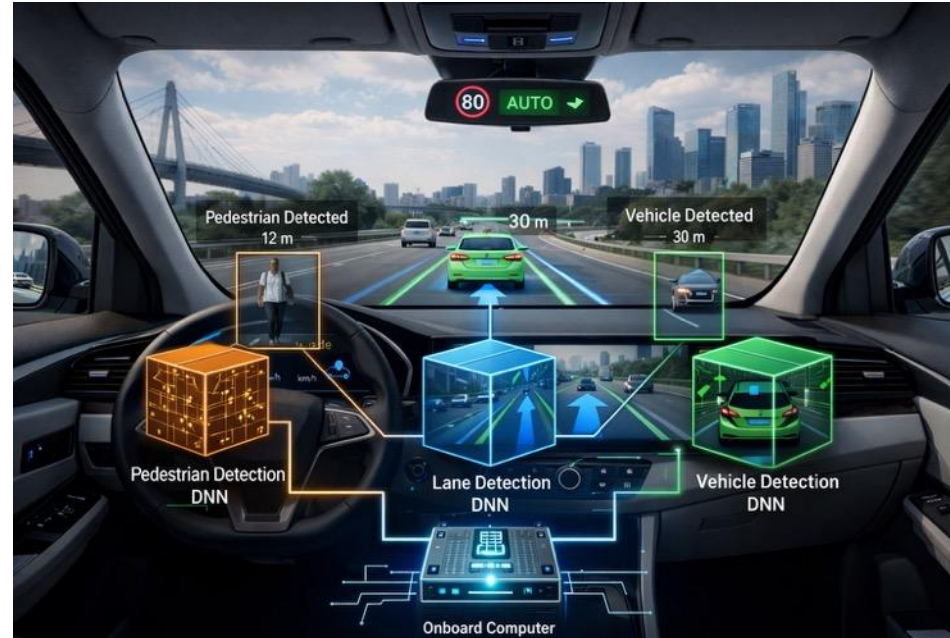
WAYNE STATE
UNIVERSITY

UNIVERSITY OF
DELAWARE



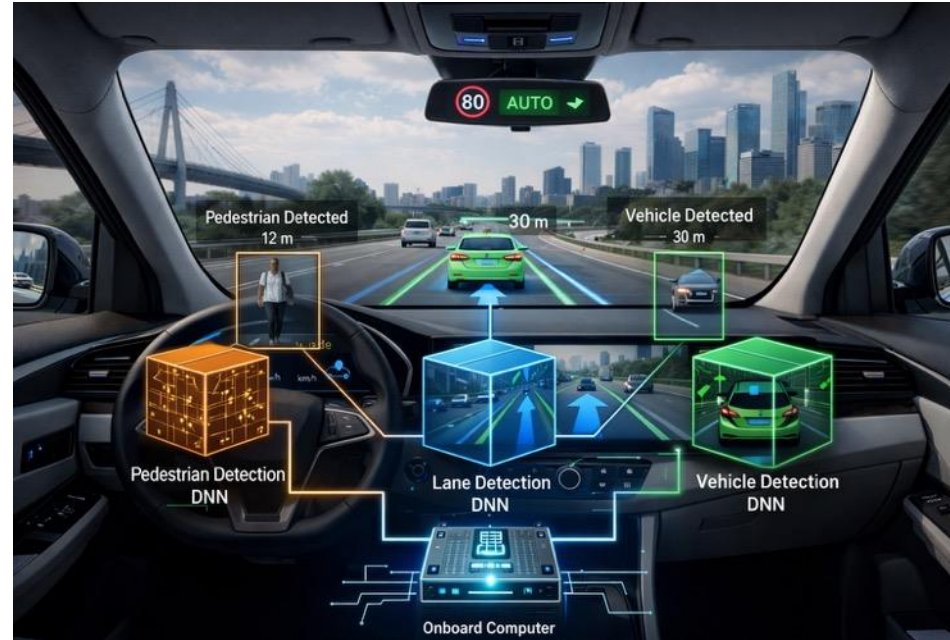
Tasks behind modern autonomous driving systems.





Tasks behind modern autonomous driving systems.





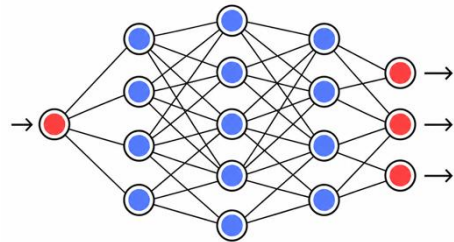
Tasks behind modern autonomous driving systems.



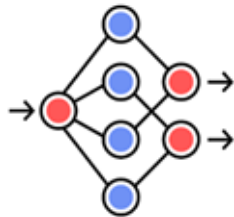
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

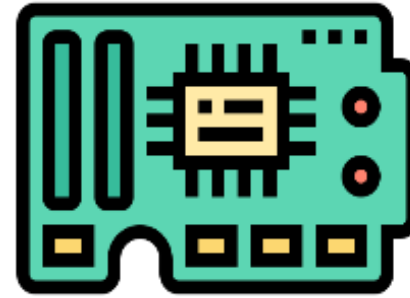
ADS applications



Large DNNs



Small DNNs

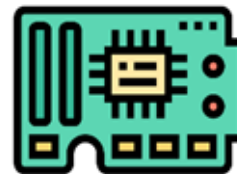


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



Accelerator for small workloads.

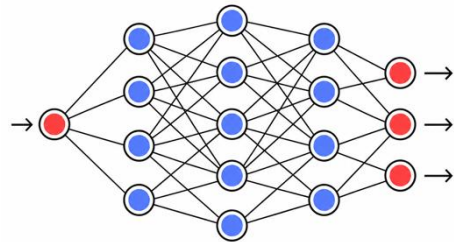
- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

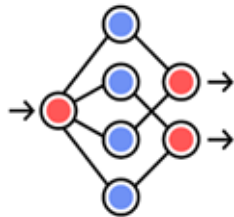
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

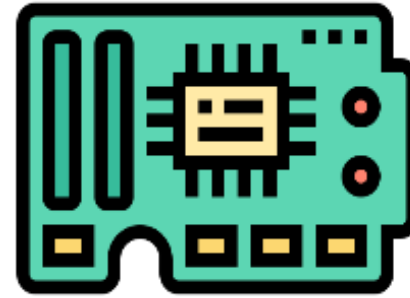
ADS applications



Large DNNs



Small DNNs

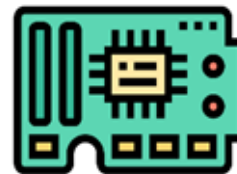


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



Accelerator for small workloads.

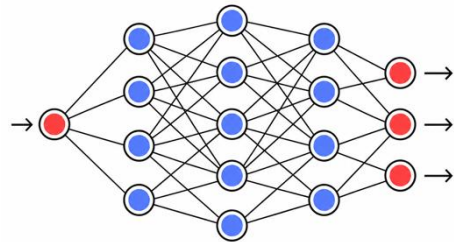
- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

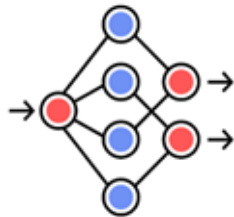
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

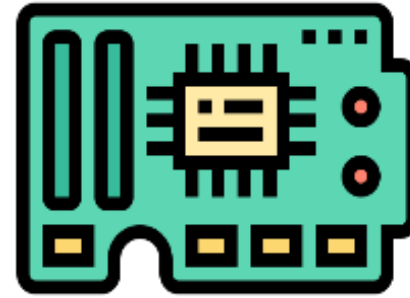
ADS applications



Large DNNs



Small DNNs

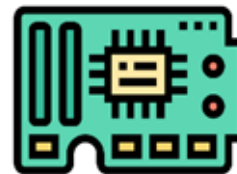


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



Accelerator for small workloads.

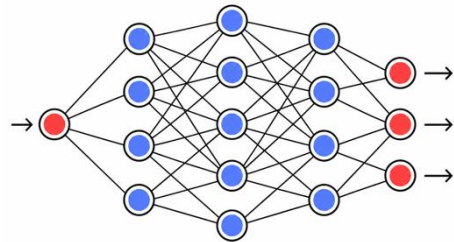
- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

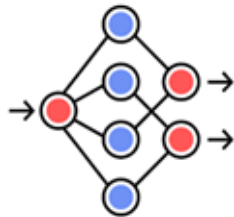
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

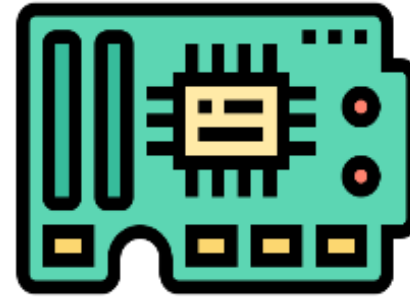
ADS applications



Large DNNs



Small DNNs

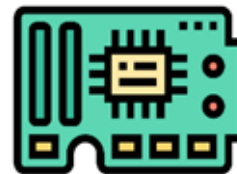


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



Accelerator for small workloads.

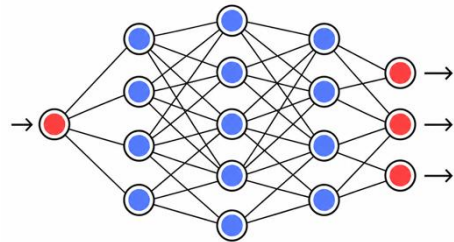
- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

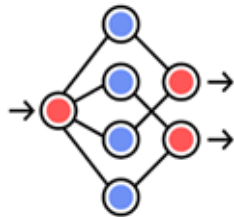
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

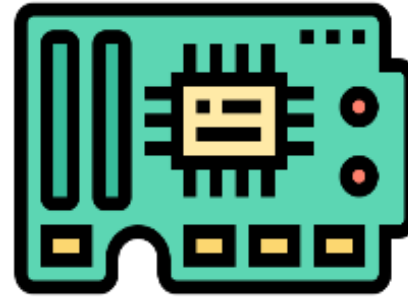
ADS applications



Large DNNs



Small DNNs

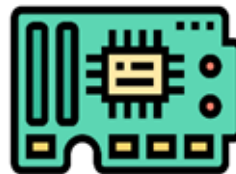


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



Accelerator for small workloads.

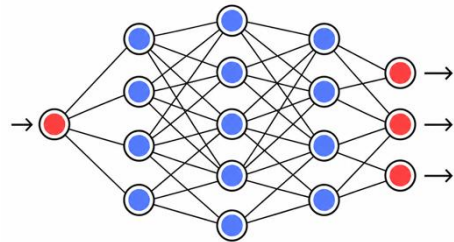
- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

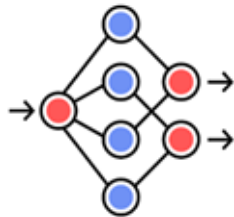
Challenge 1

- Autonomous driving system requires **diverse** DNN models.

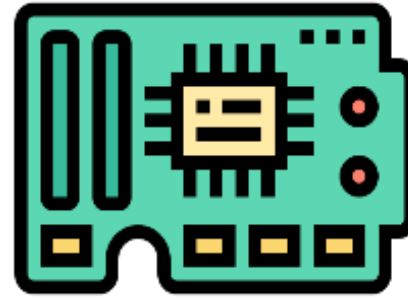
ADS applications



Large DNNs



Small DNNs

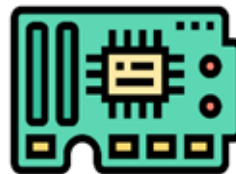


Accelerator for large workloads.

- Good for large DNNs;
- Bad for small DNNs.



Switching bitstream requires ~20ms.



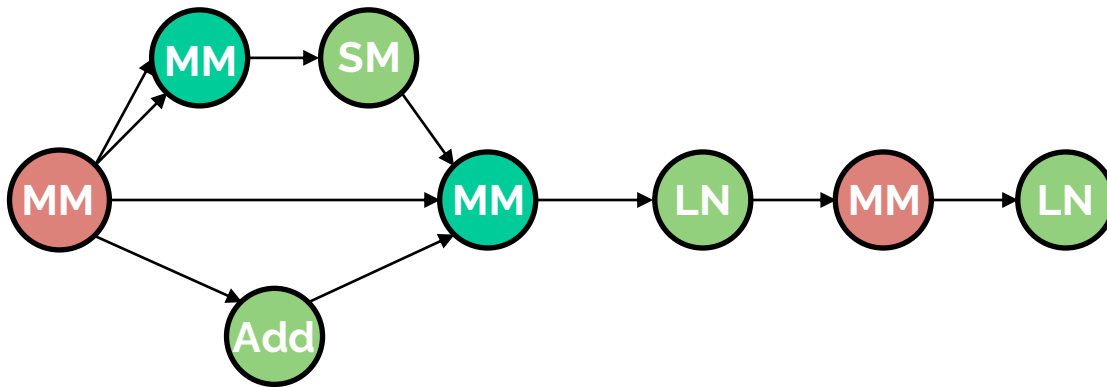
Accelerator for small workloads.

- Bad for large DNNs;
- Good for small DNNs.

- How to **efficiently** handle diverse DNN models in **one bitstream**?

Challenge 2

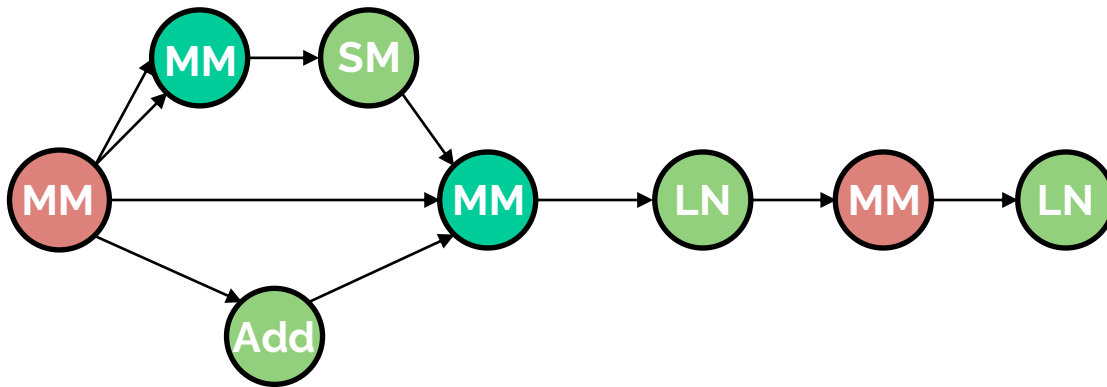
- Complicated applications lead to a **huge design space**.
- The design space is exponential to # of layers, which is $O(m^n)$.
 - m is **# of mapping solutions** for a single layer.
 - n is **# of layers**.



950x The U.S. GDP in 2025.

Challenge 2

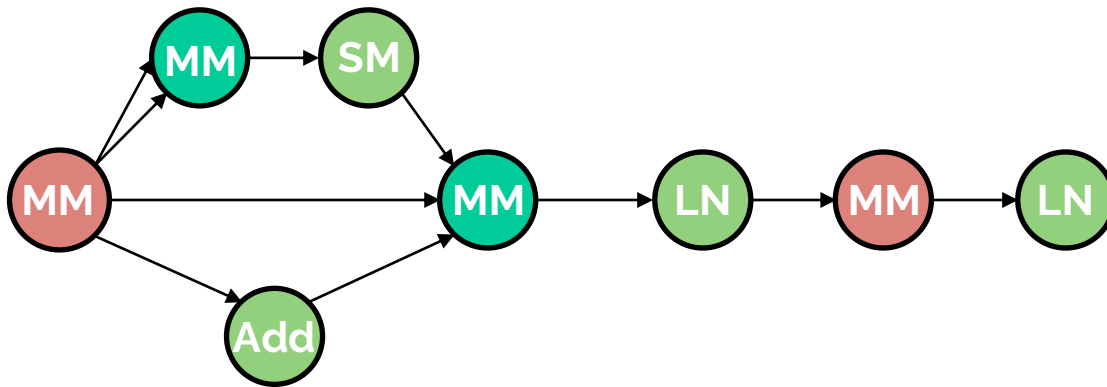
- Complicated applications lead to a **huge design space**.
- The design space is exponential to # of layers, which is $O(m^n)$.
 - m is **# of mapping solutions** for a single layer.
 - n is **# of layers**.



950x The U.S. GDP in 2025.

Challenge 2

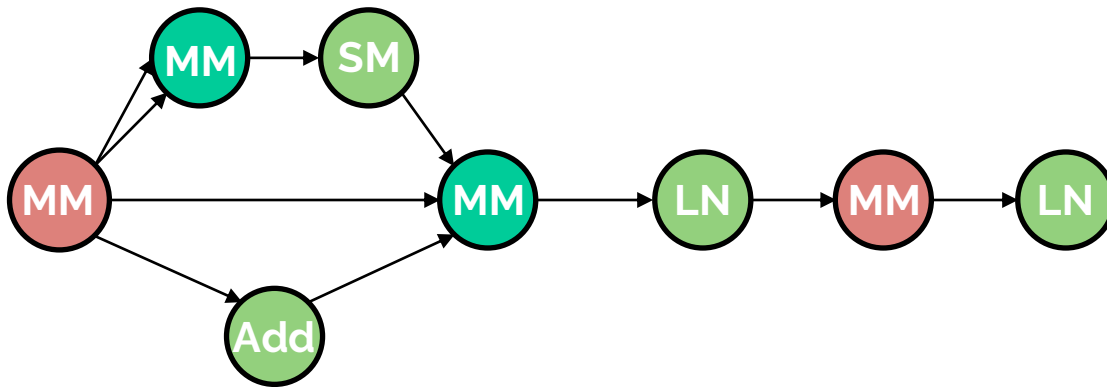
- Complicated applications lead to a **huge design space**.
- The design space is exponential to # of layers, which is $O(m^n)$.
 - m is **# of mapping solutions** for a single layer.
 - n is **# of layers**.



950x The U.S. GDP in 2025.

Challenge 2

- Complicated applications lead to a **huge design space**.
- The design space is exponential to # of layers, which is $O(m^n)$.
 - m is **# of mapping solutions** for a single layer.
 - n is **# of layers**.



950x The U.S. GDP in 2025.

Challenge 3

- Given a task set, automatic framework requires balancing the **optimality** and **scalability**.
 - Brute-force search induces redundant search time.
 - Heuristic search loses many opportunities for better mapping solutions.
- How to maintain the **optimality** while preserving the **efficiency**?

Challenge 3

- Given a task set, automatic framework requires balancing the **optimality** and **scalability**.
 - Brute-force search induces redundant search time.
 - Heuristic search loses many opportunities for better mapping solutions.
- How to maintain the **optimality** while preserving the **efficiency**?

Challenge 3

- Given a task set, automatic framework requires balancing the **optimality** and **scalability**.
 - Brute-force search induces redundant search time.
 - Heuristic search loses many opportunities for better mapping solutions.
- How to maintain the **optimality** while preserving the **efficiency**?

Challenge 3

- Given a task set, automatic framework requires balancing the **optimality** and **scalability**.
 - Brute-force search induces redundant search time.
 - Heuristic search loses many opportunities for better mapping solutions.
- How to maintain the **optimality** while preserving the **efficiency**?

DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Challenges

- Diverse operations exist in modern DNN models.
- Complicated applications lead to a huge design space.
- Balance between optimality and scalability.

Solutions

- **DORA proposes an overlay architecture as well as an ISA abstraction for diverse workloads.**
- **DORA is equipped with an automatic framework that can analyze the mapping solutions at a fine-grained layer level.**
- **DORA has a two-stage design space exploration that combines brute-force and heuristic search.**

DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Challenges

- Diverse operations exist in modern DNN models.
- Complicated applications lead to a huge design space.
- Balance between optimality and scalability.

Solutions

- **DORA proposes an overlay architecture as well as an ISA abstraction for diverse workloads.**
- **DORA is equipped with an automatic framework that can analyze the mapping solutions at a fine-grained layer level.**
- **DORA has a two-stage design space exploration that combines brute-force and heuristic search.**

DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Challenges

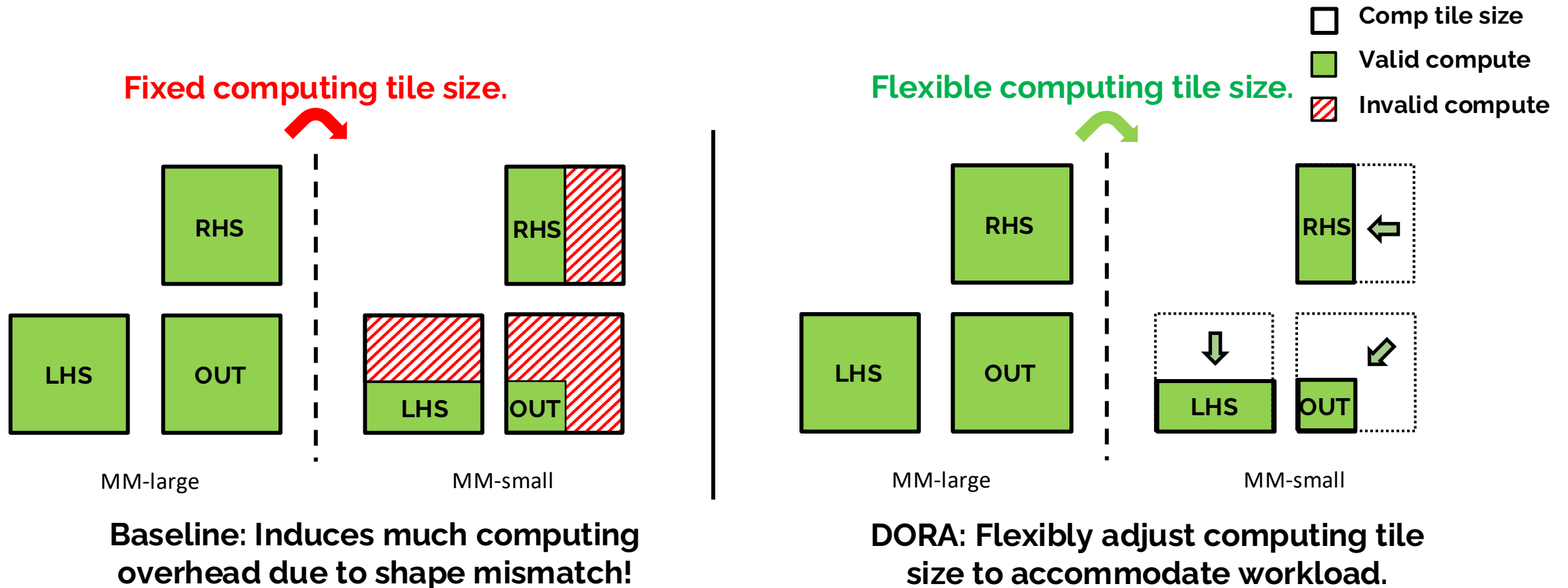
- Diverse operations exist in modern DNN models.
- Complicated applications lead to a huge design space.
- Balance between optimality and scalability.

Solutions

- **DORA proposes an overlay architecture as well as an ISA abstraction for diverse workloads.**
- **DORA is equipped with an automatic framework that can analyze the mapping solutions at a fine-grained layer level.**
- **DORA has a two-stage design space exploration that combines brute-force and heuristic search.**

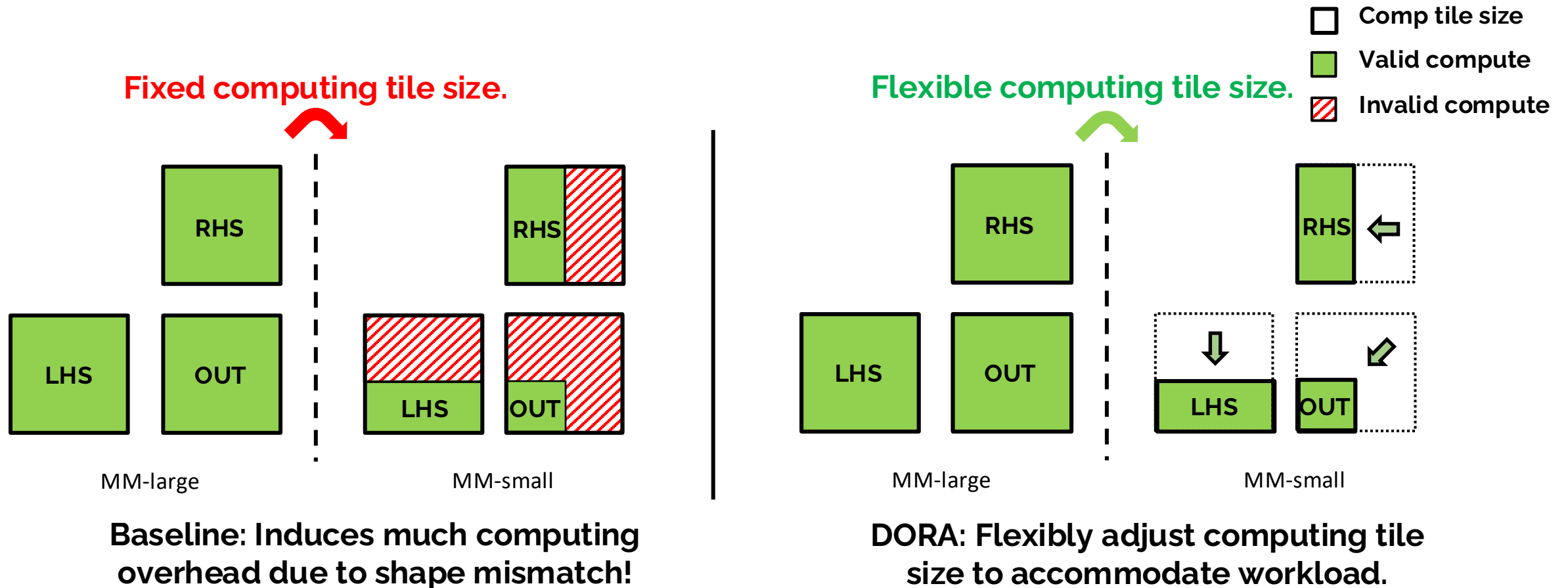
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



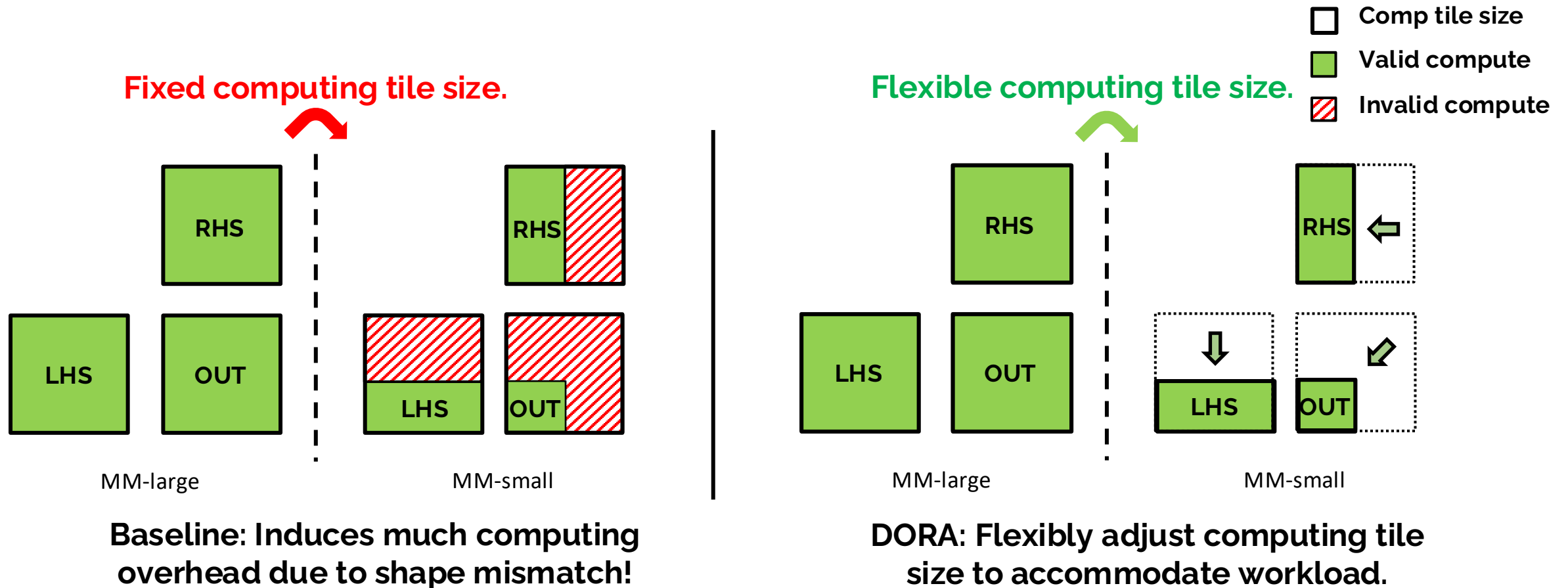
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



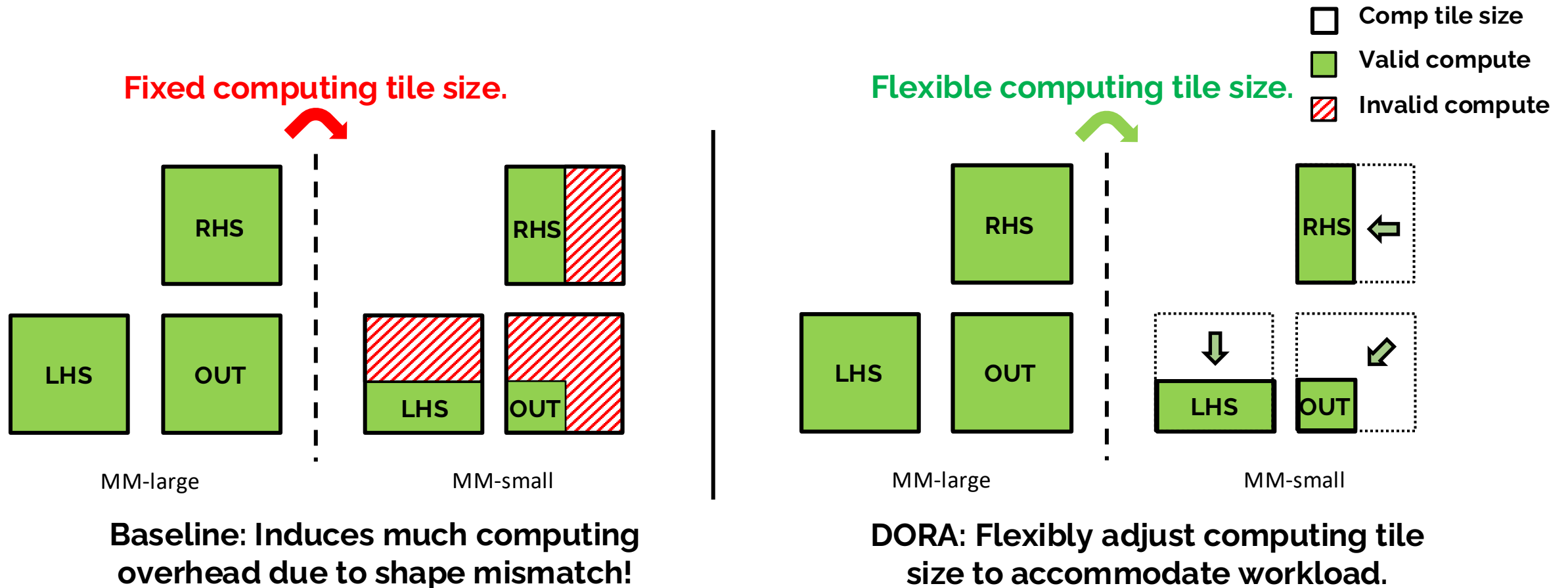
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



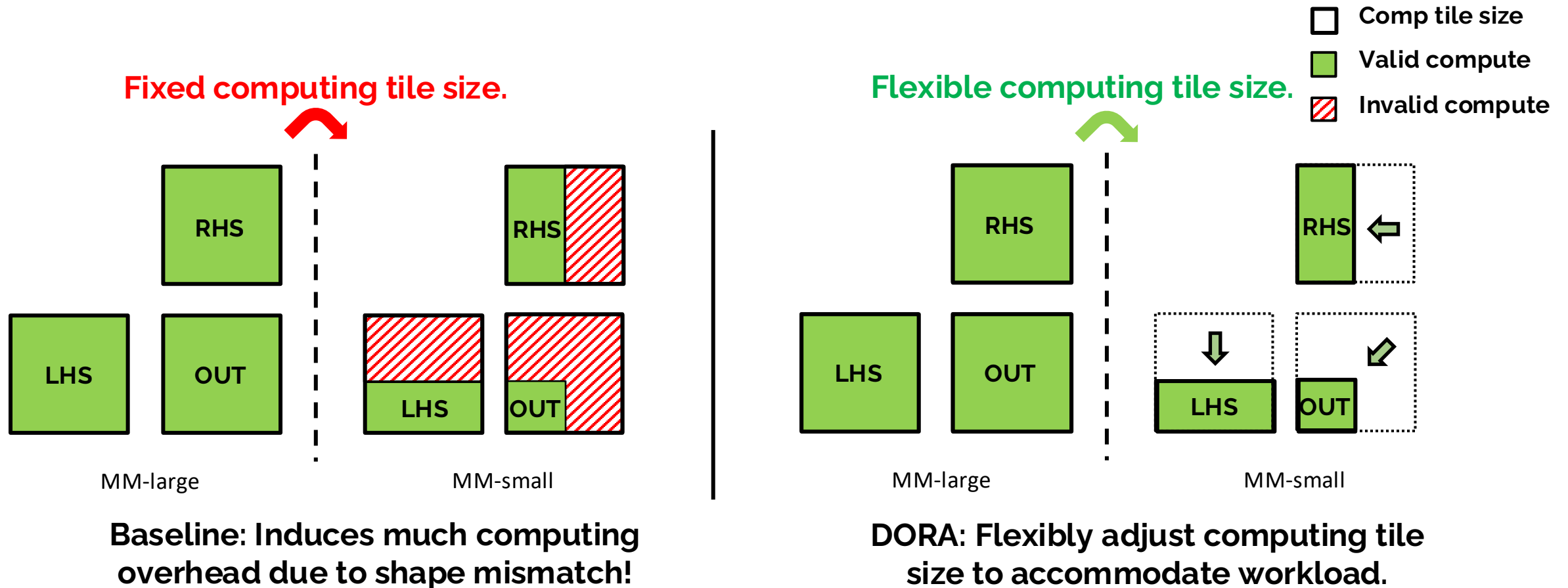
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



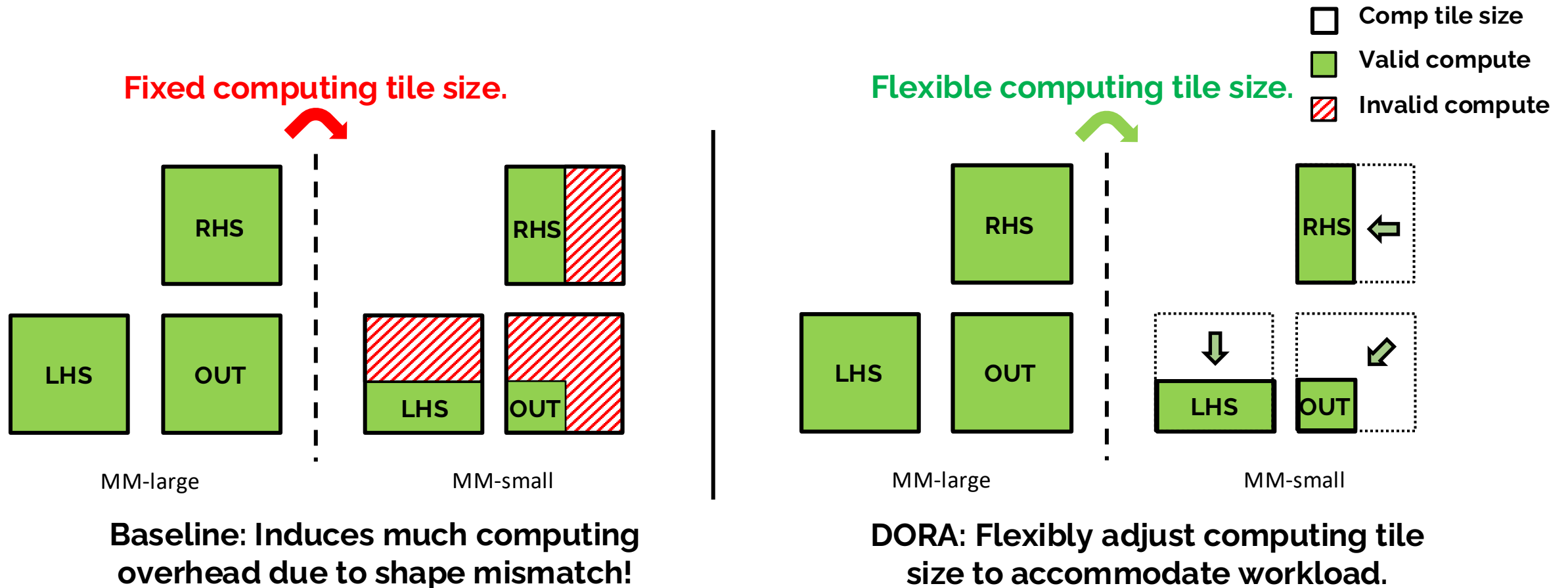
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



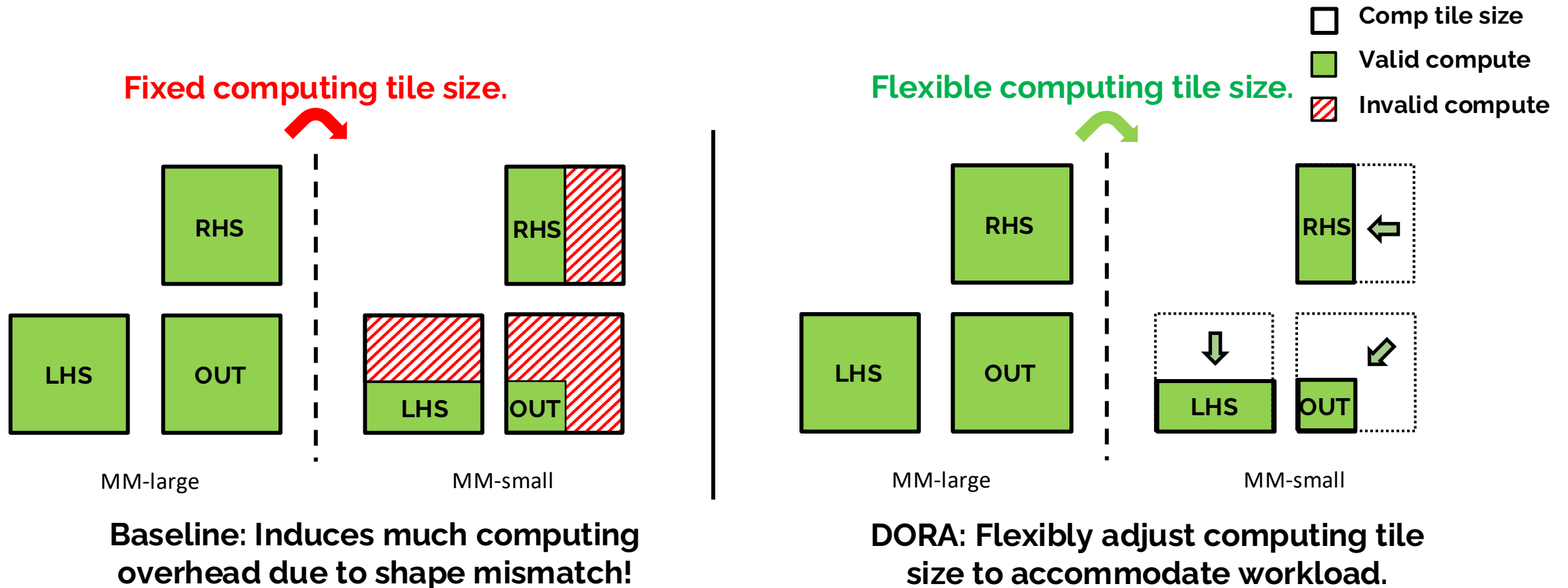
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



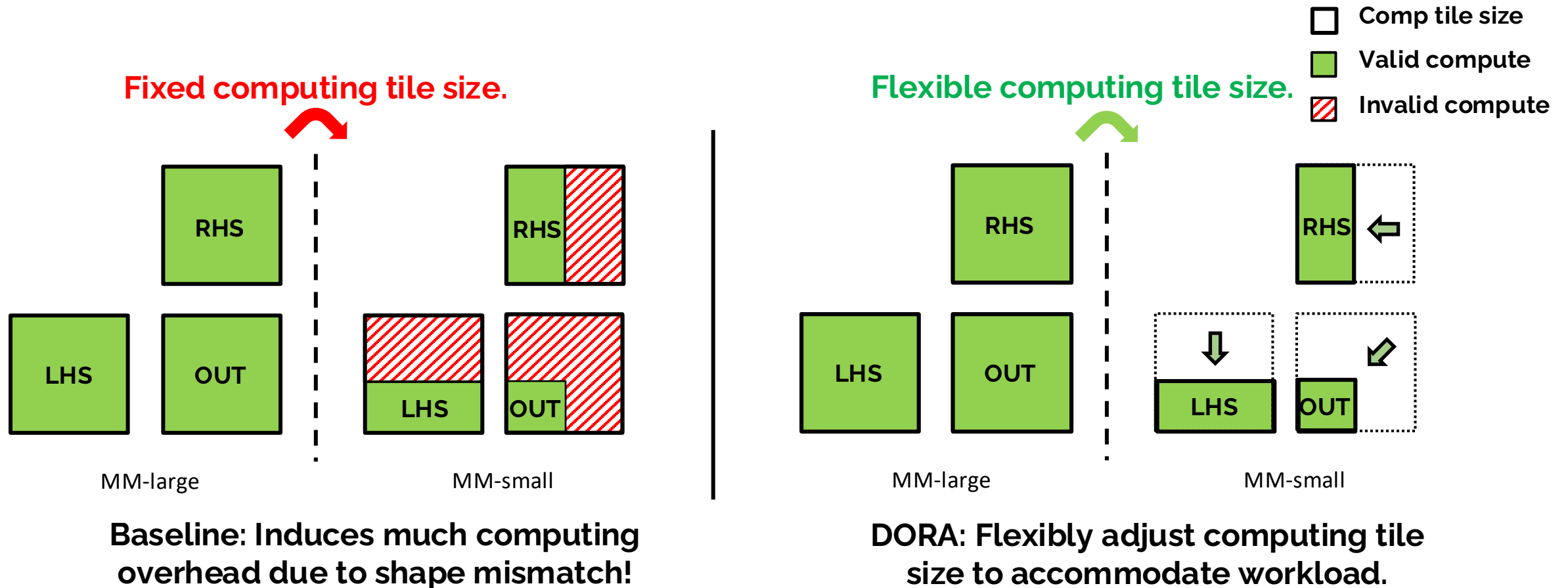
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



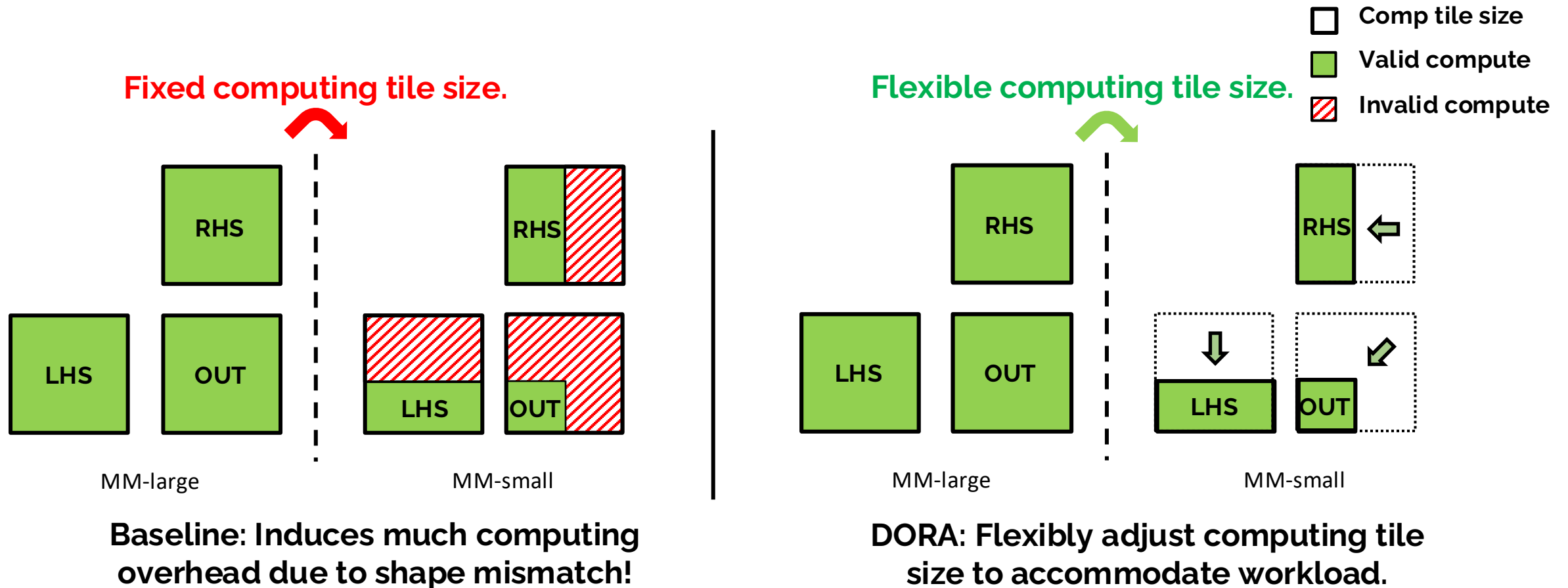
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



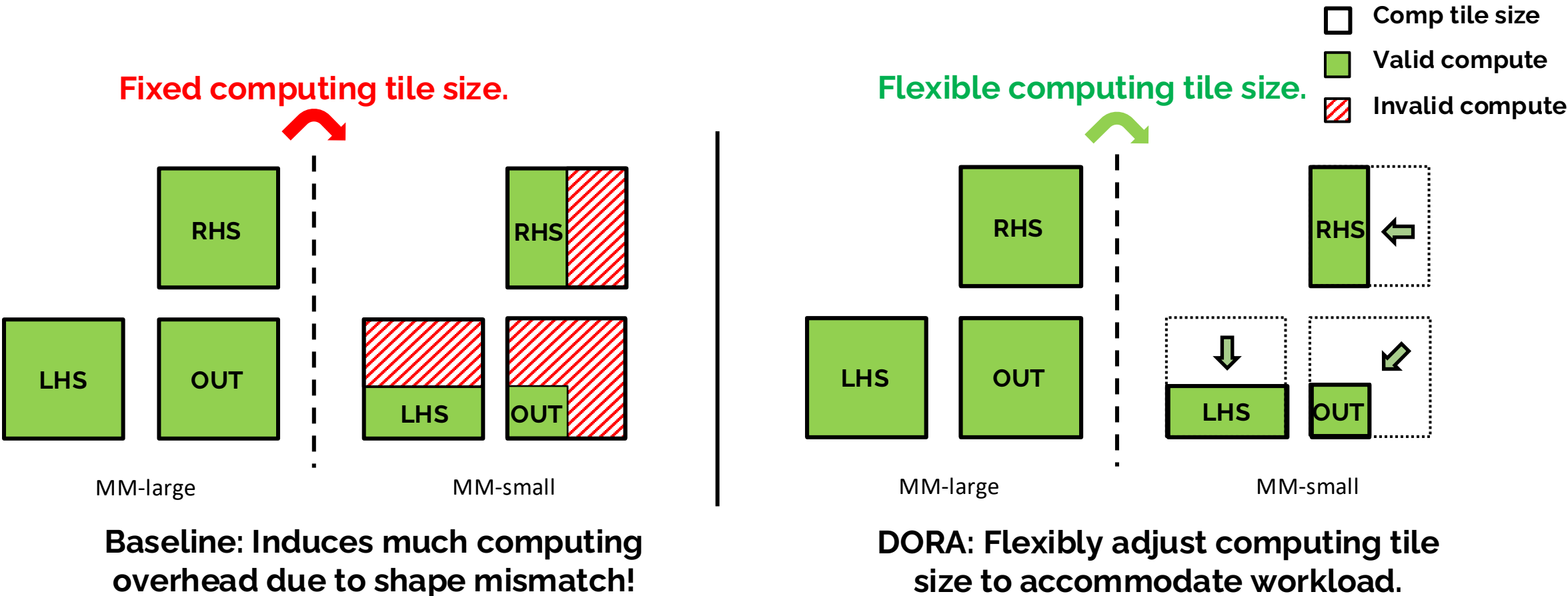
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



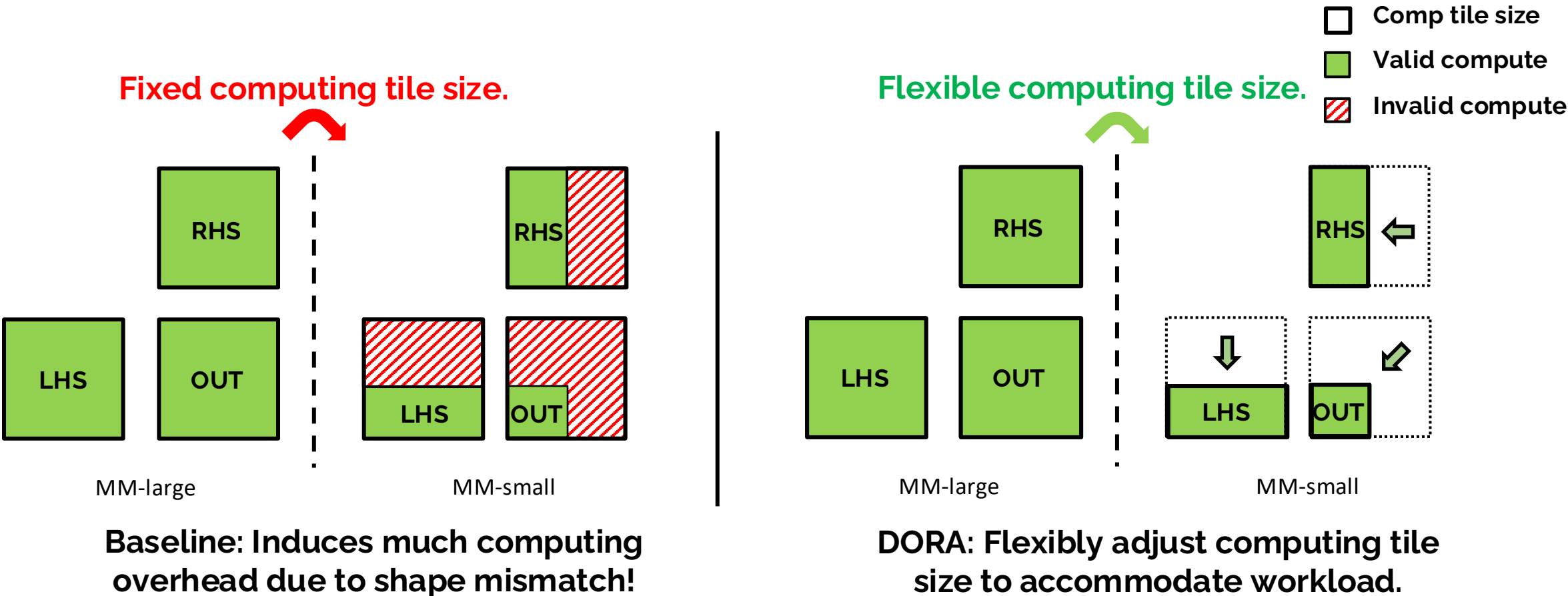
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



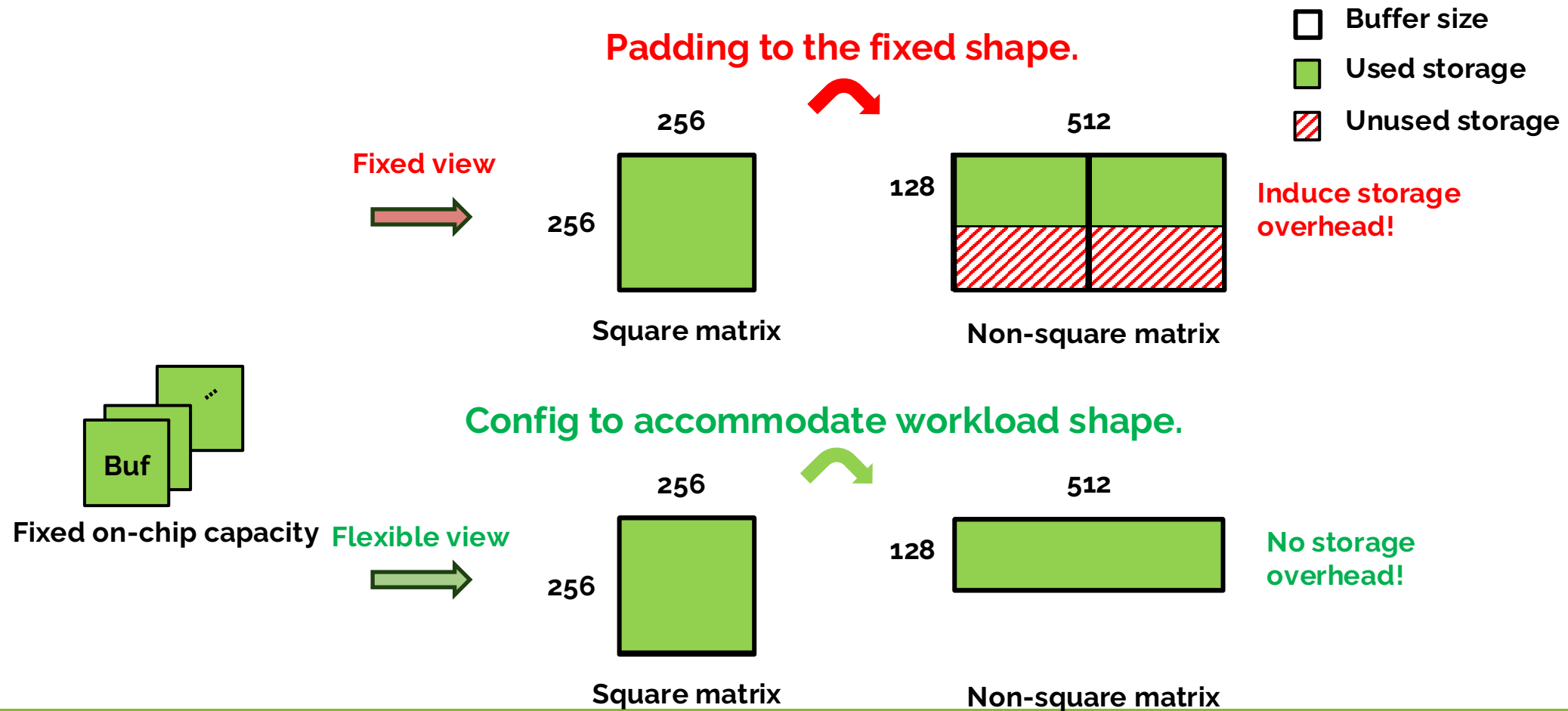
DORA Architecture Design

- Enable flexible **computation tile size** by **parallelism management**.



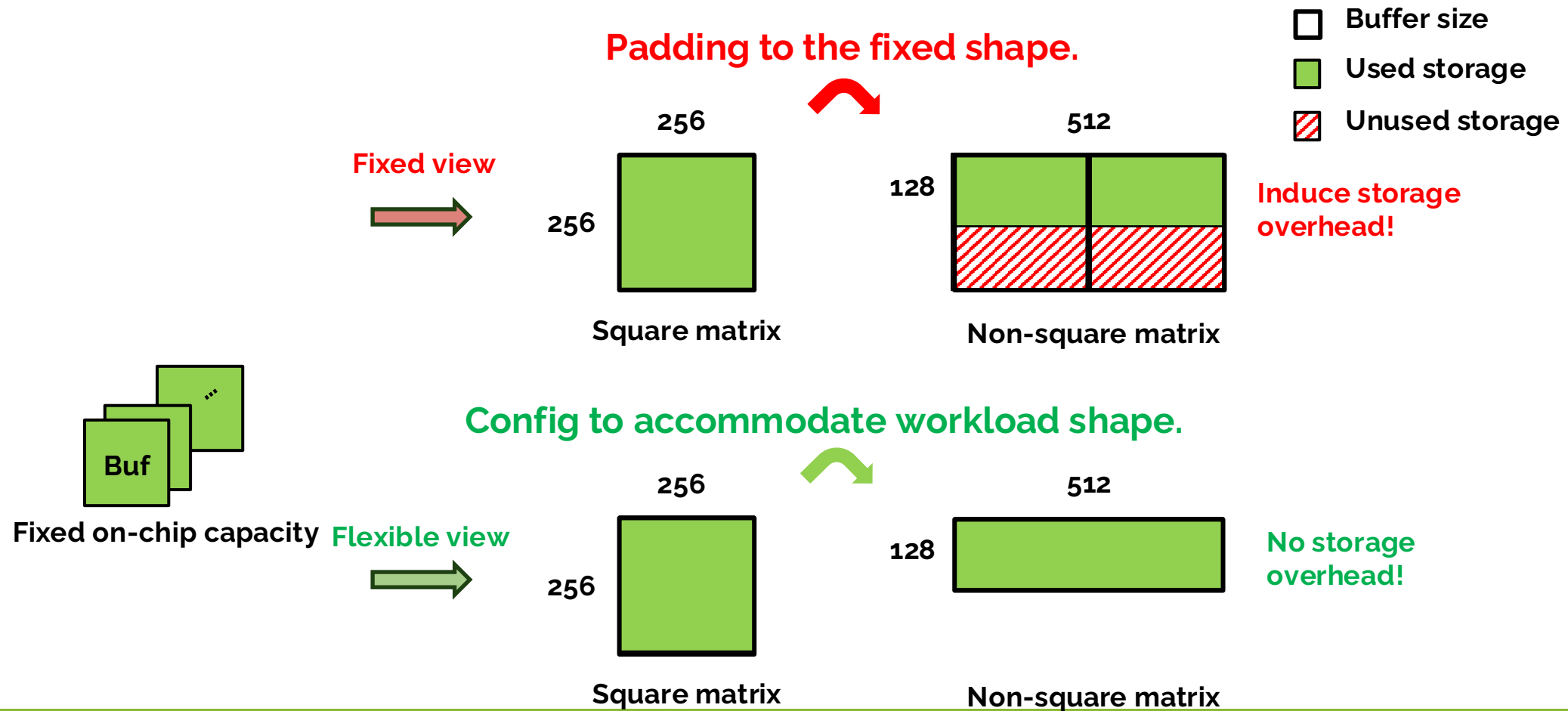
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



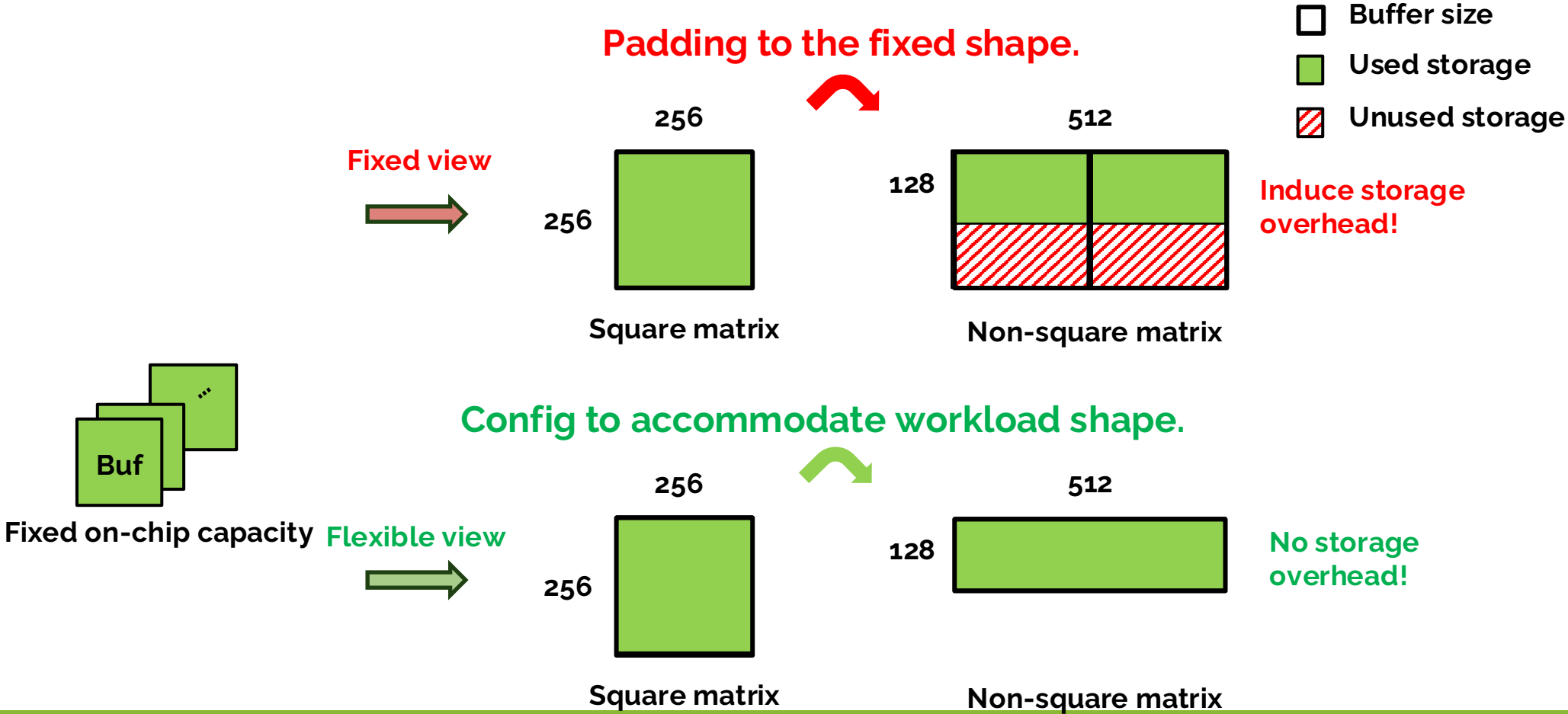
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



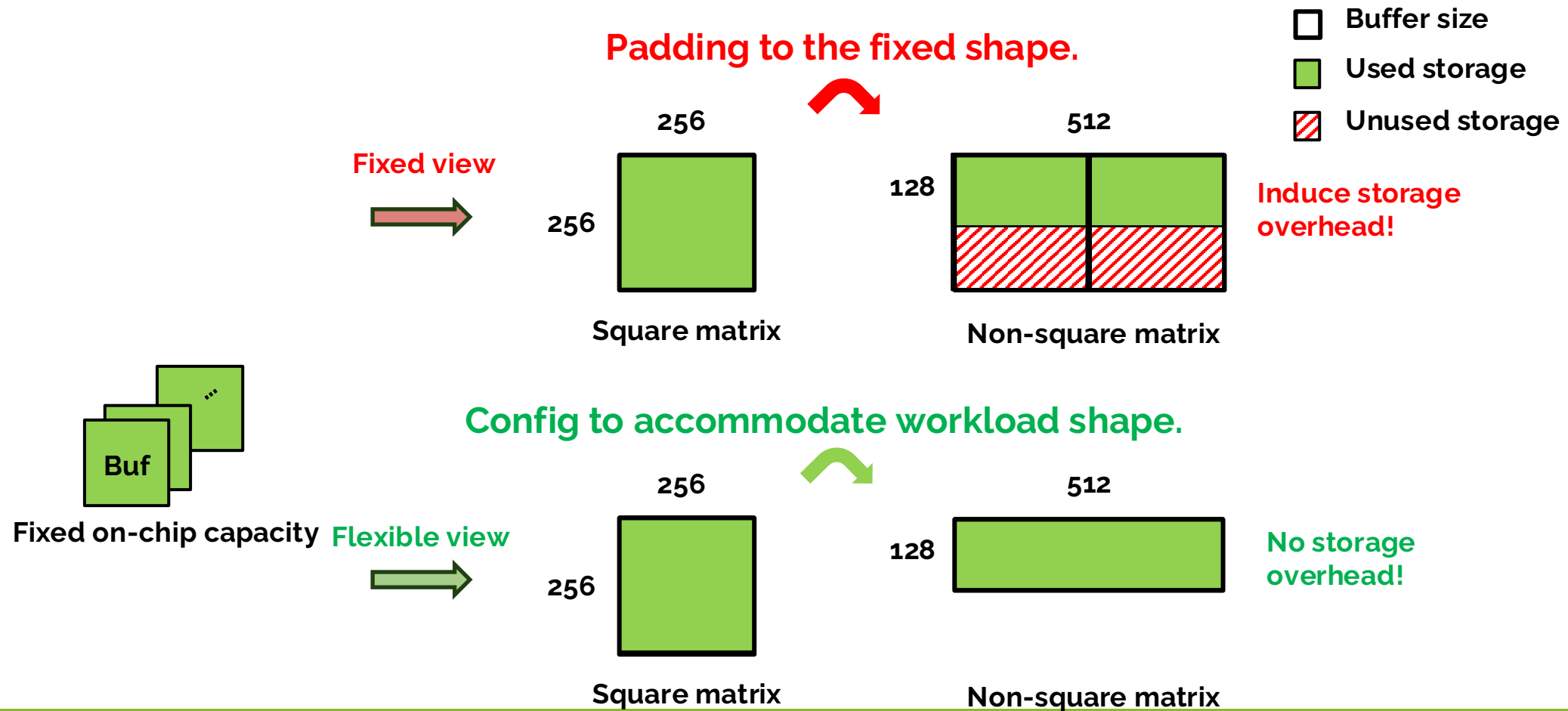
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



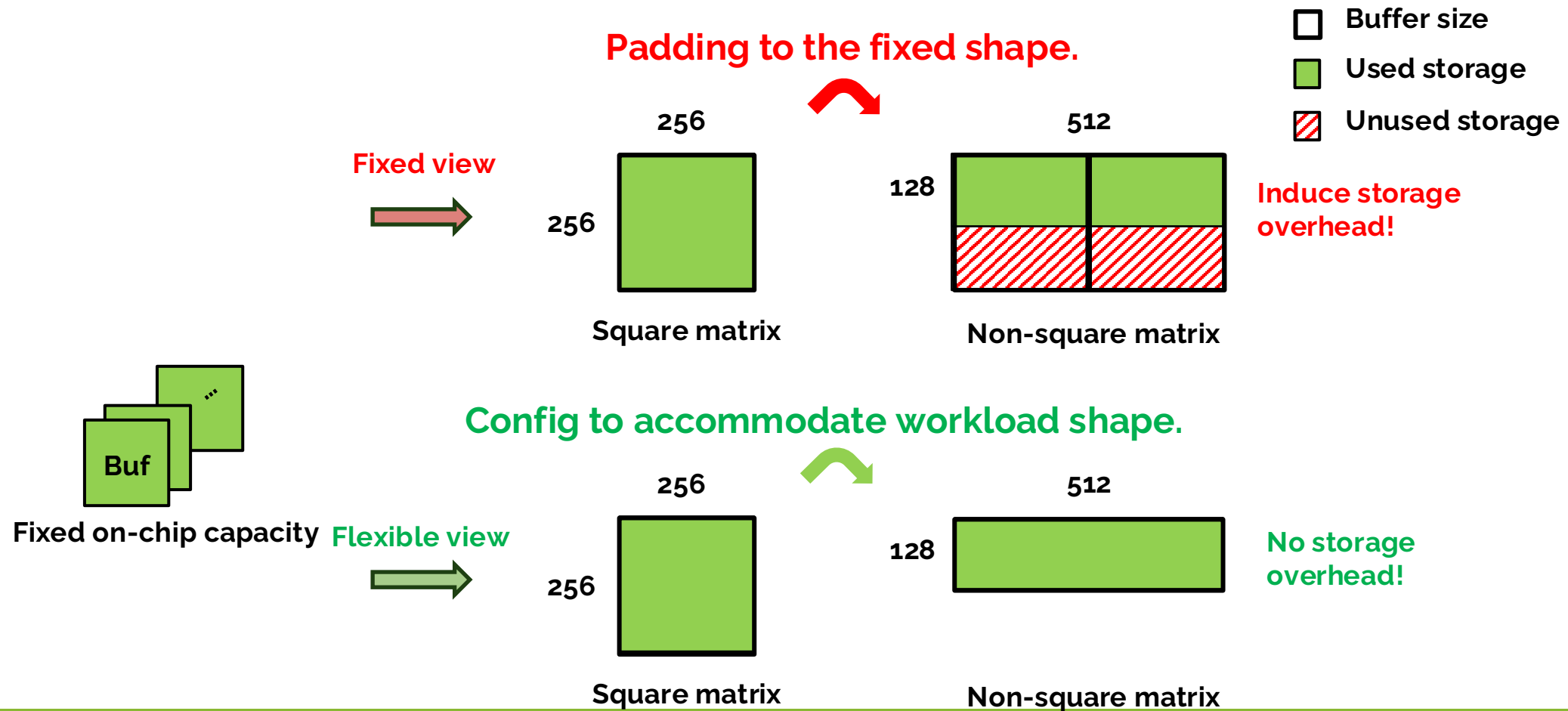
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



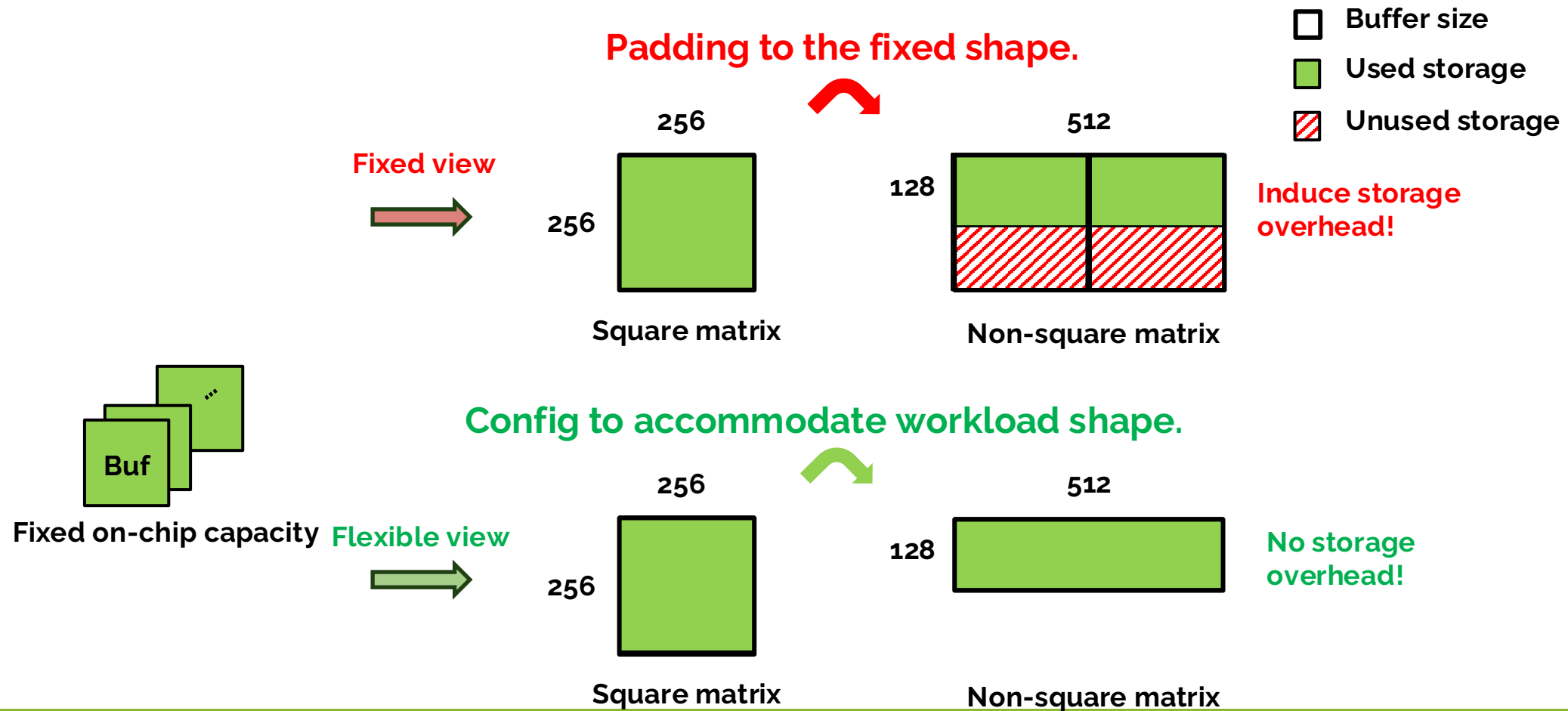
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



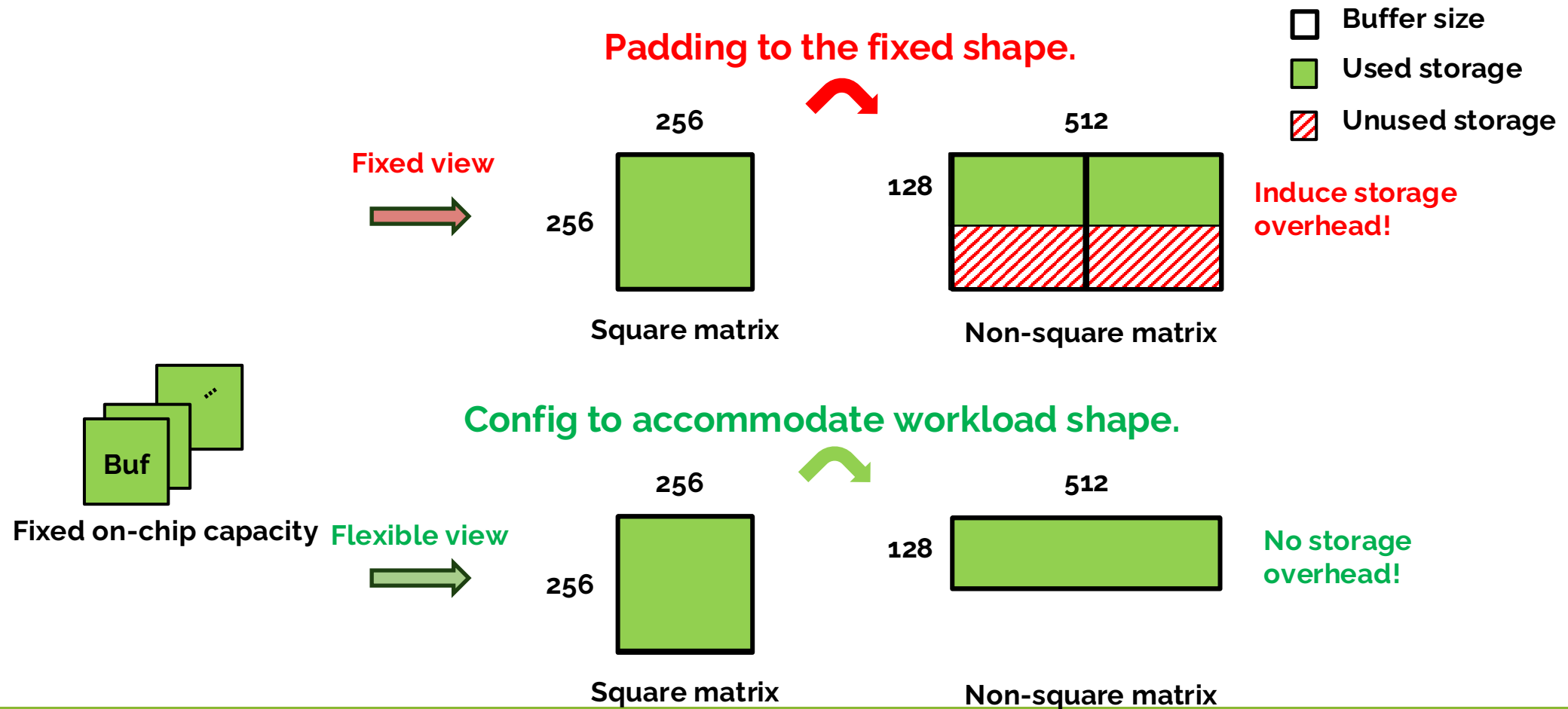
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



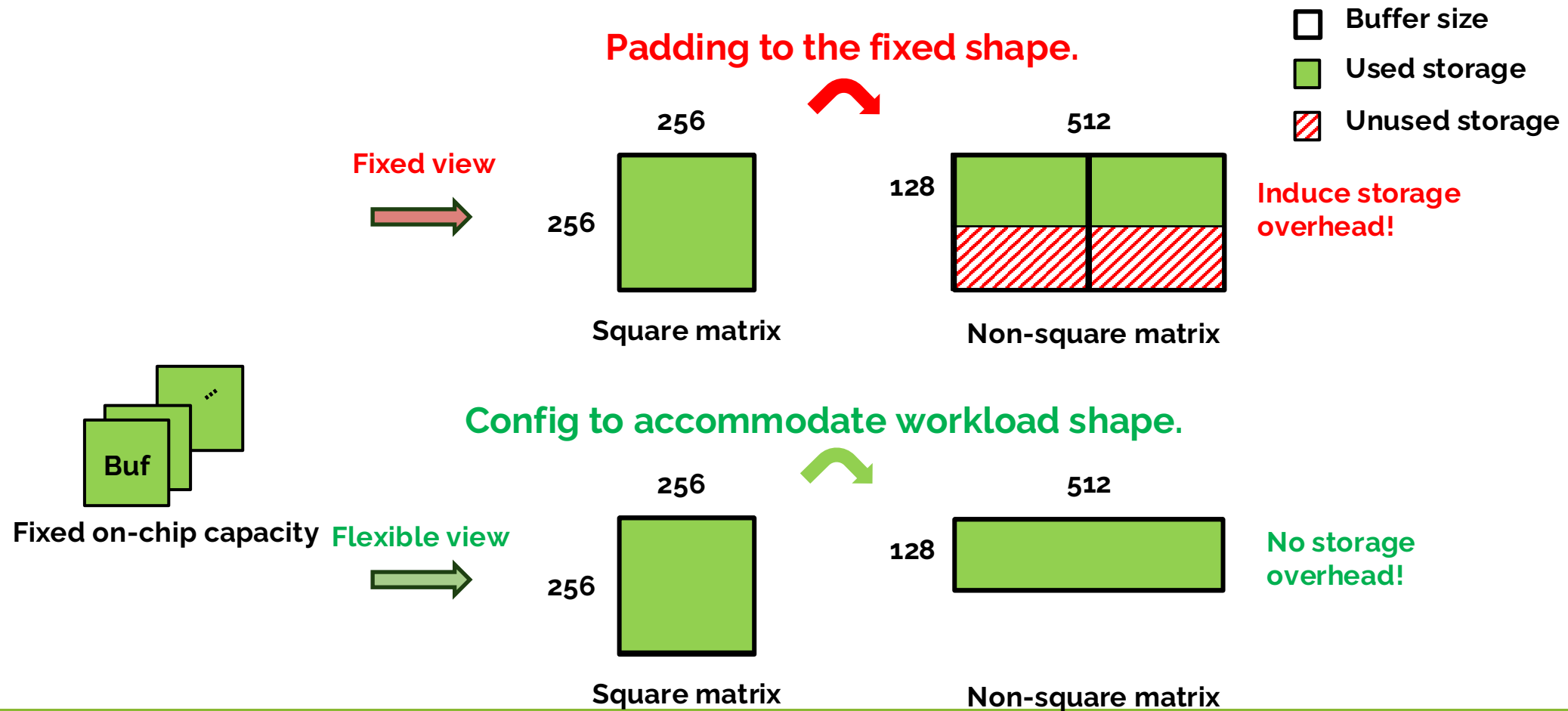
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



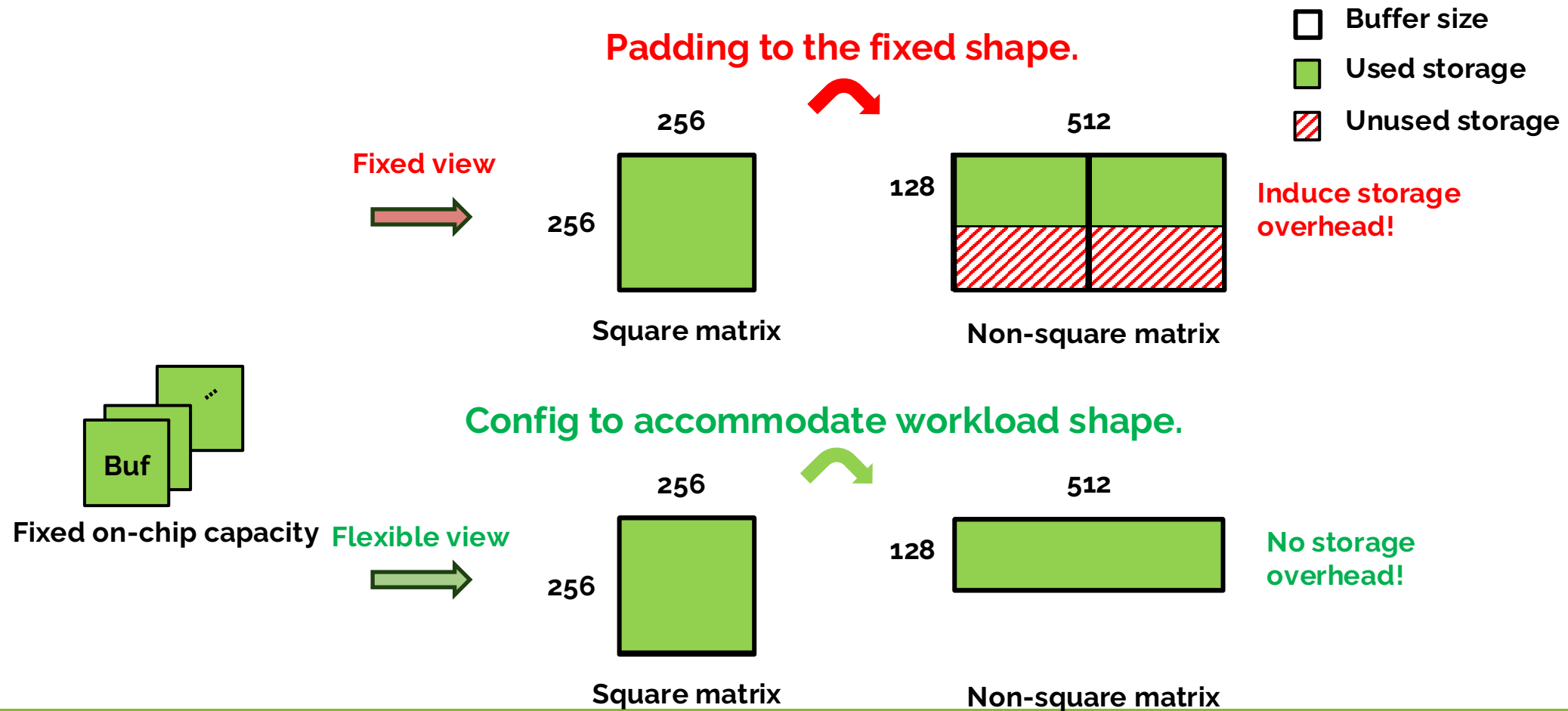
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



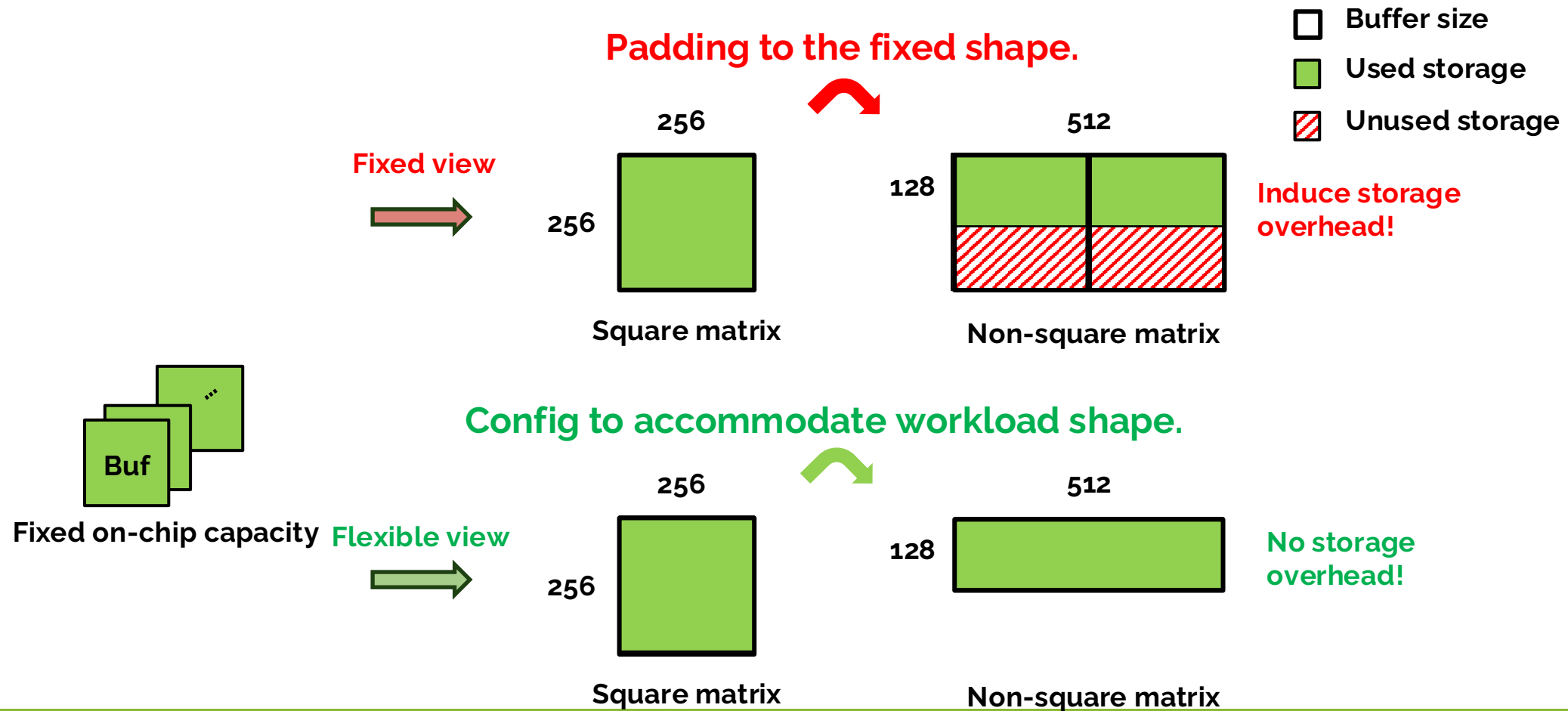
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



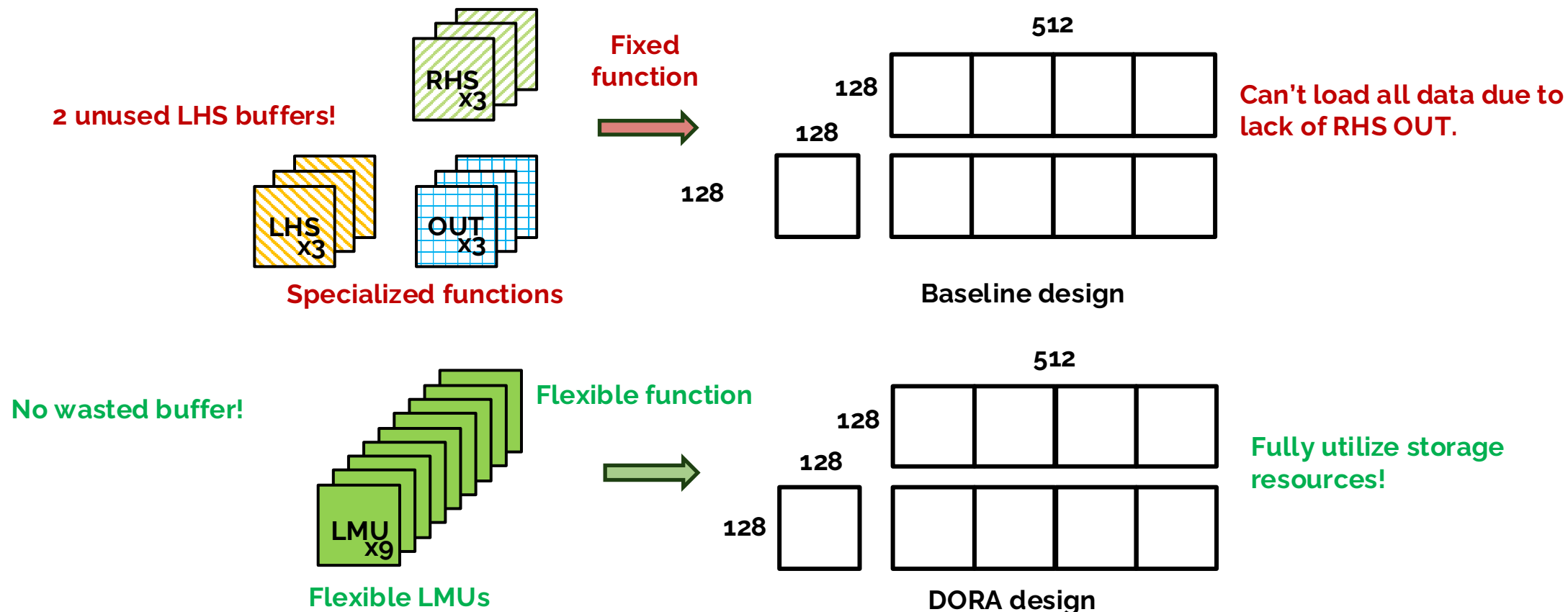
DORA Architecture Design

- Enable flexible **memory view** by on-chip **buffer management**.



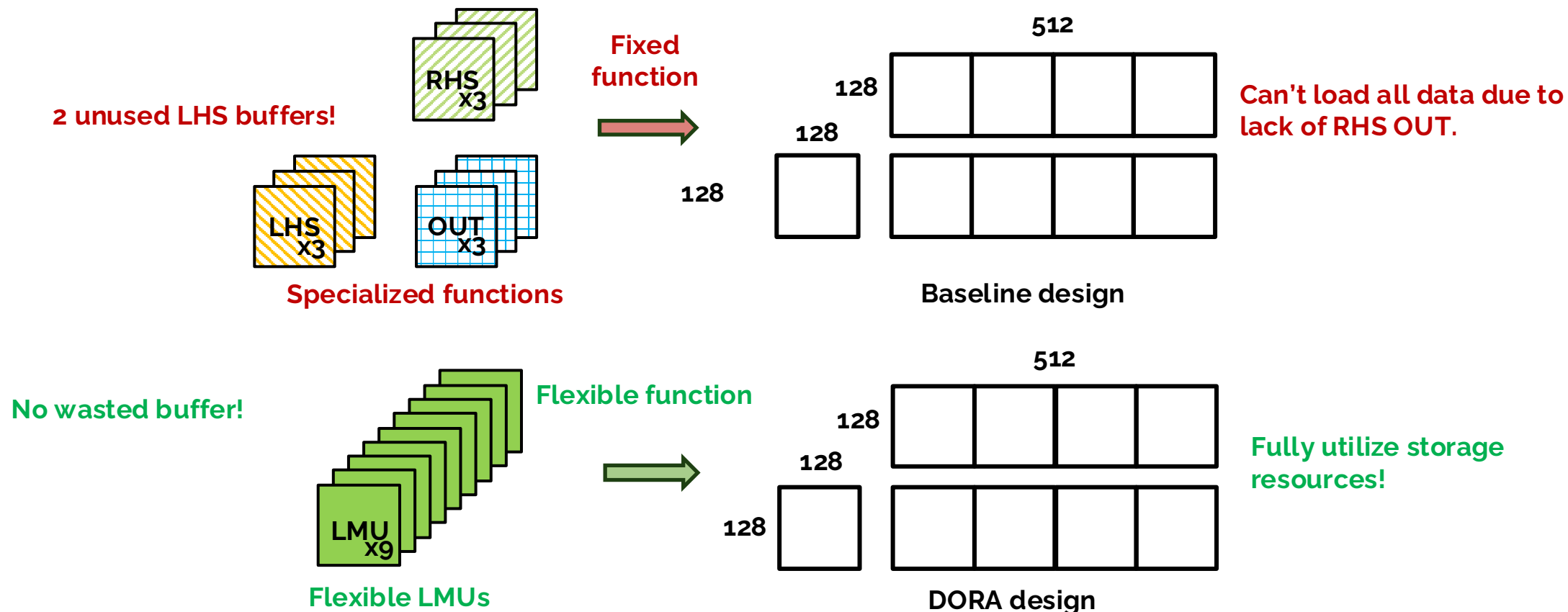
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



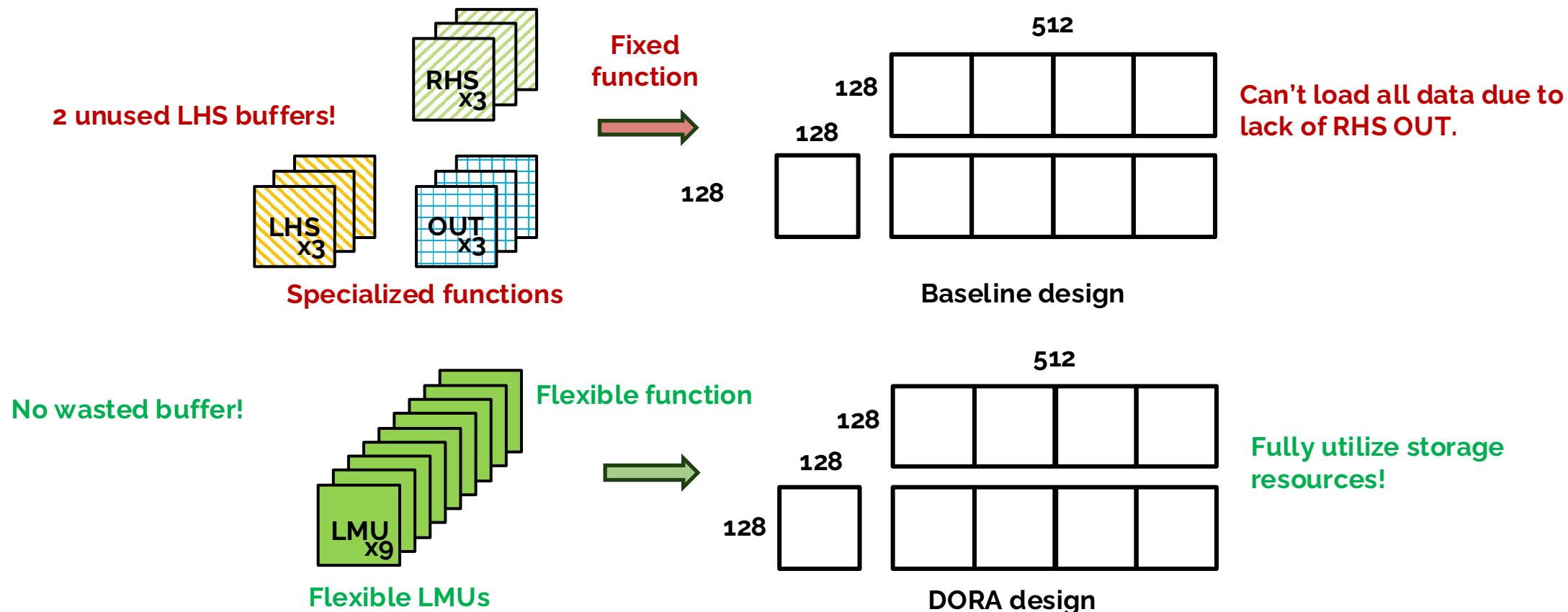
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



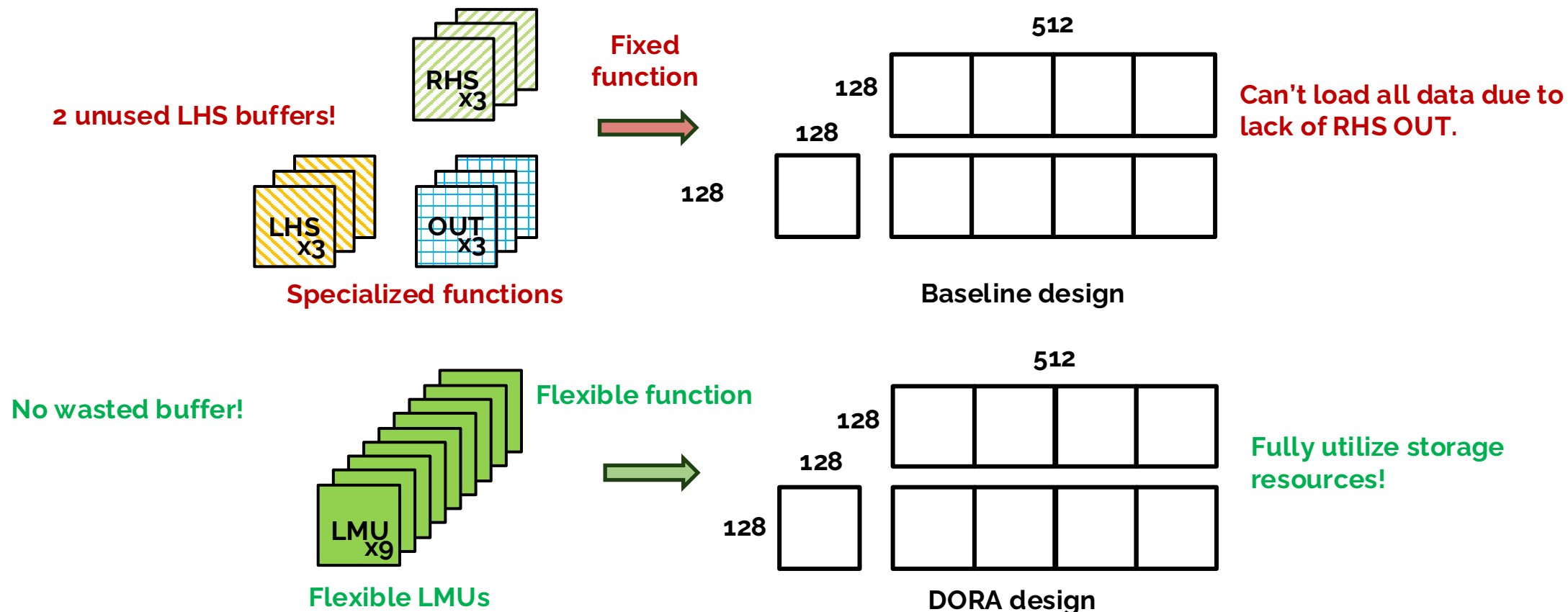
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



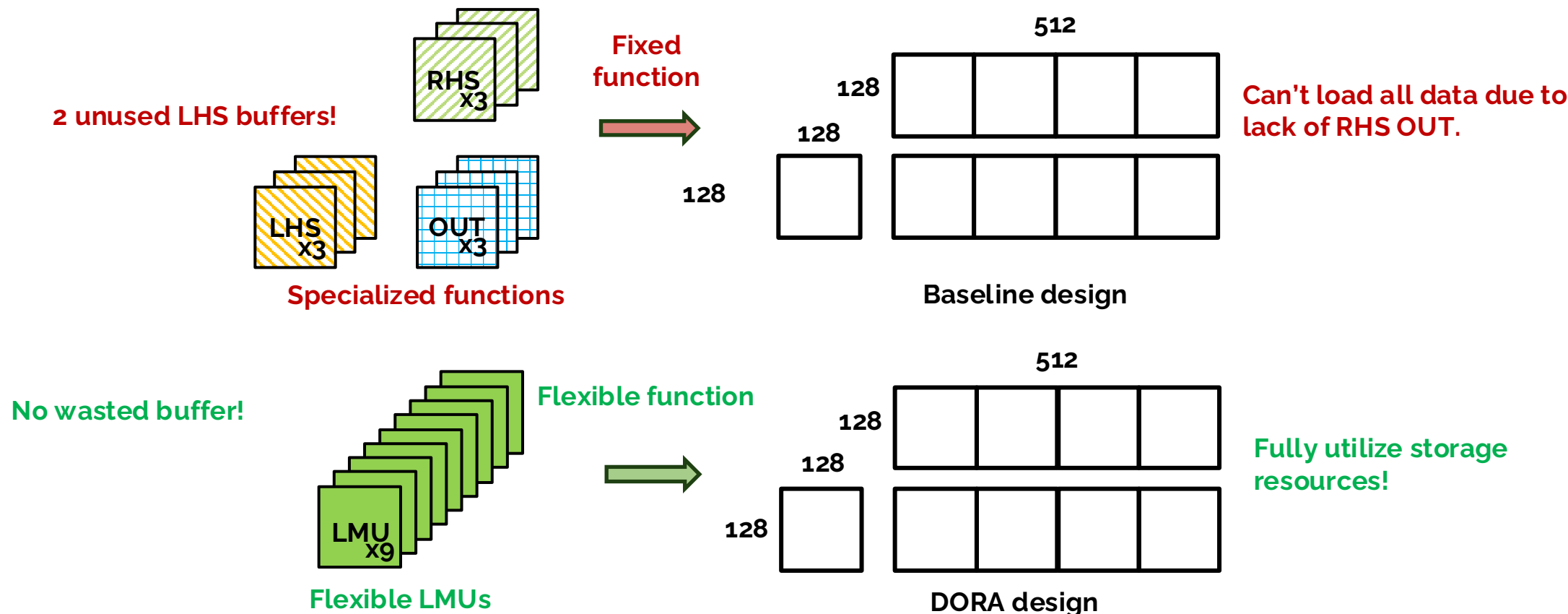
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



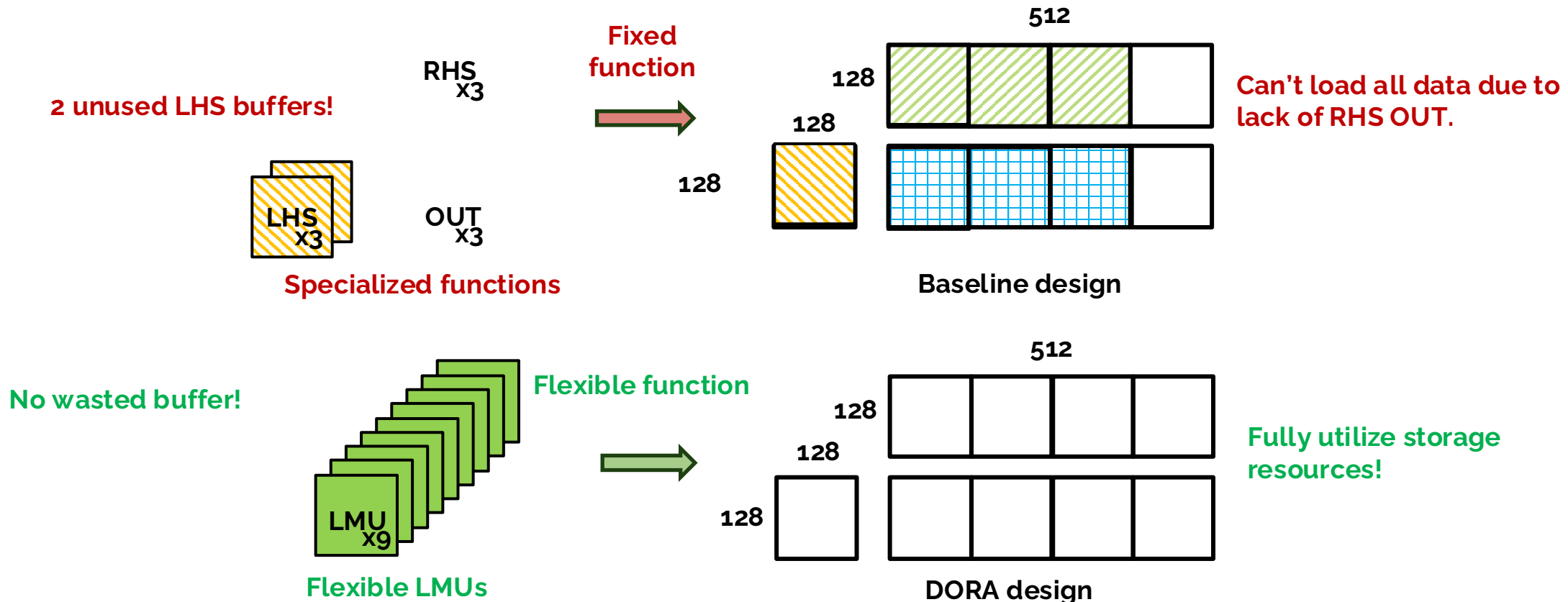
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



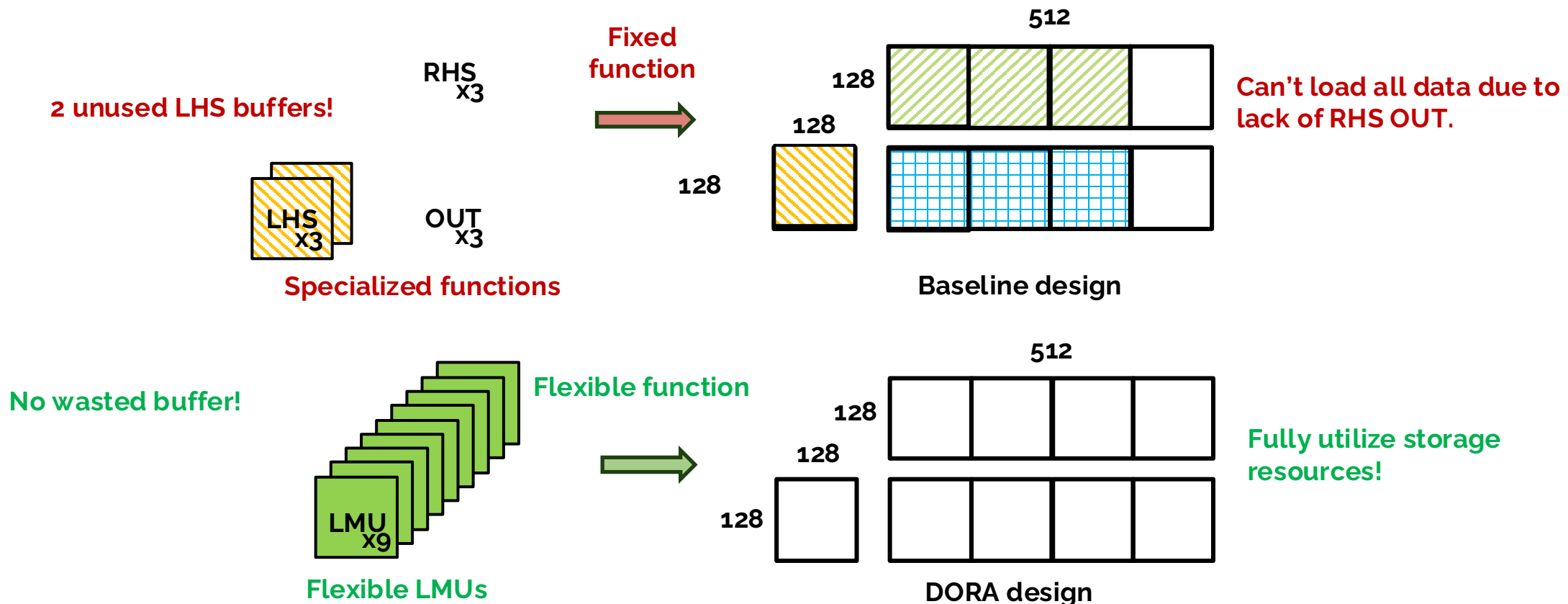
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



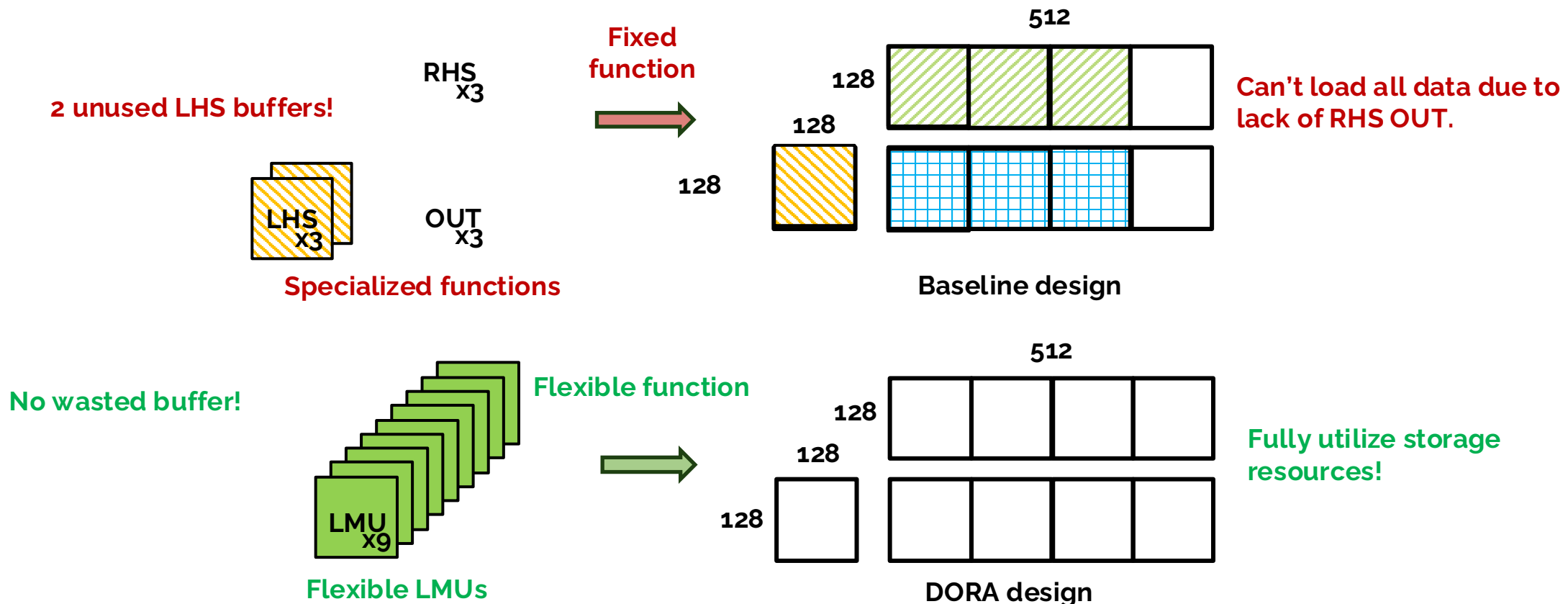
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



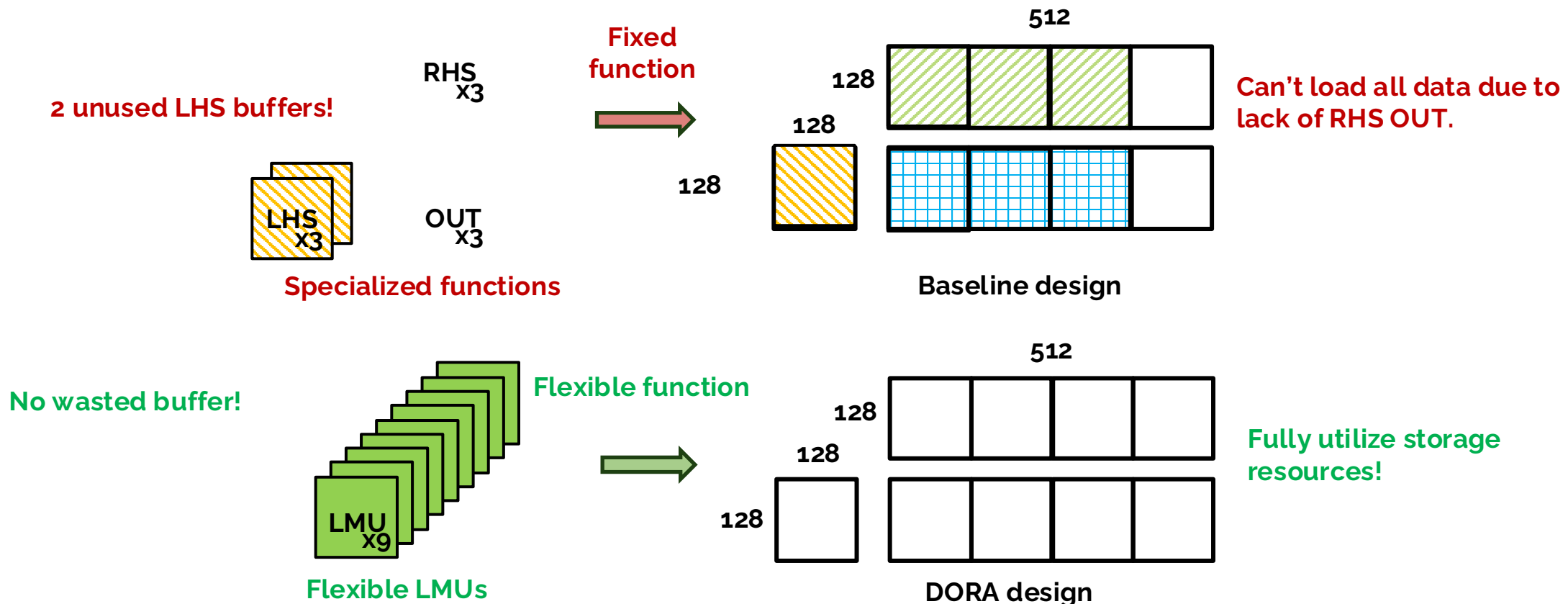
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



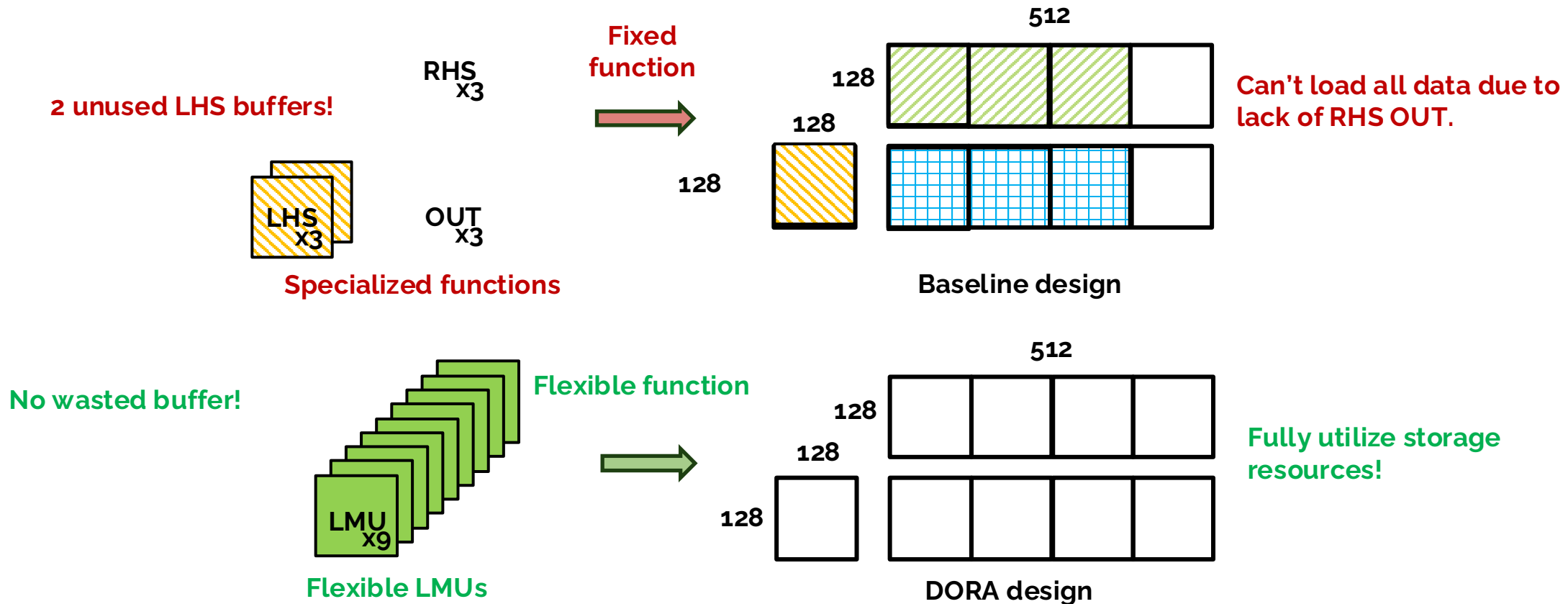
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



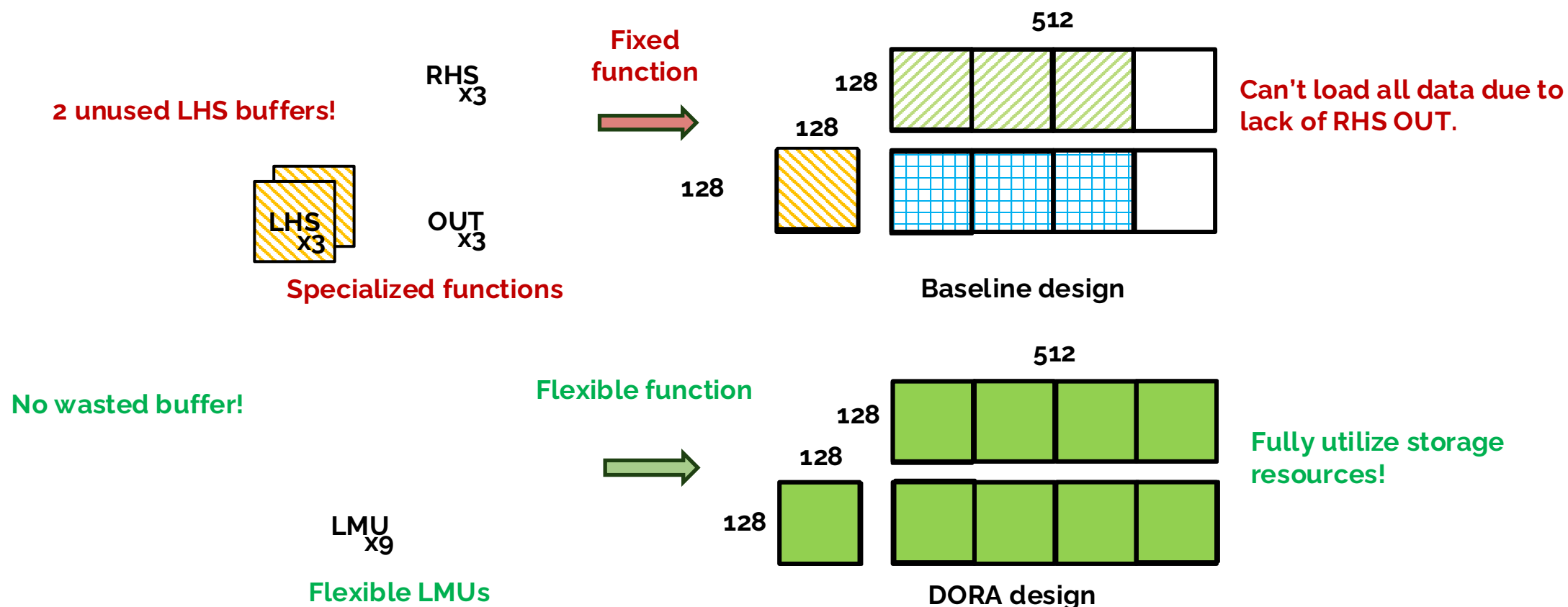
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



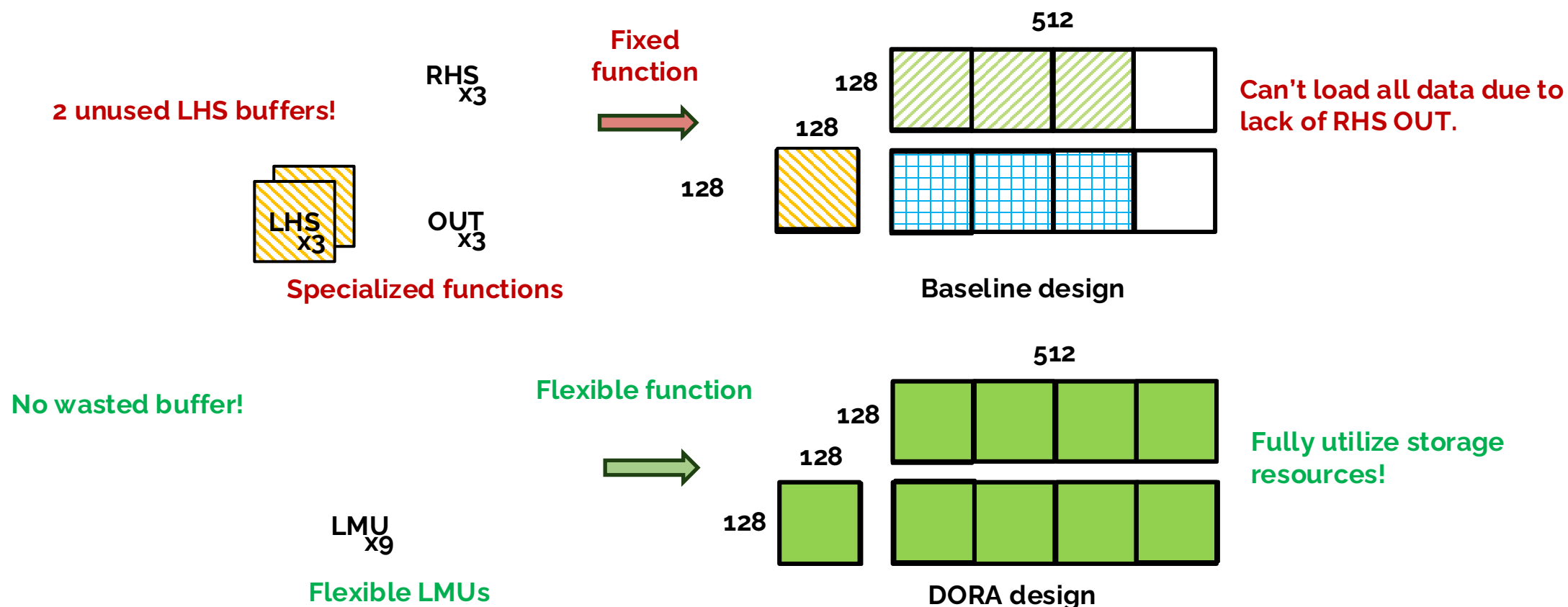
DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



DORA Architecture Design

- Enable flexible **memory functionalities** by **topology management**.



DORA Architecture Design

- DORA overlay can orchestrate **dataflow** with **instructions** to enable flexible execution patterns for **diverse workloads**.

How to **efficiently** program DORA overlay for **software developers**?



DORA Architecture Design

How to **efficiently** program DORA overlay for **software developers**?



DORA Architecture Design

- DORA overlay can orchestrate **dataflow** with **instructions** to enable flexible execution patterns for **diverse workloads**.

How to **efficiently** program DORA overlay for **software developers**?



DORA Architecture Design

- DORA overlay can orchestrate **dataflow** with **instructions** to enable flexible execution patterns for **diverse workloads**.

How to **efficiently** program DORA overlay for **software developers**?



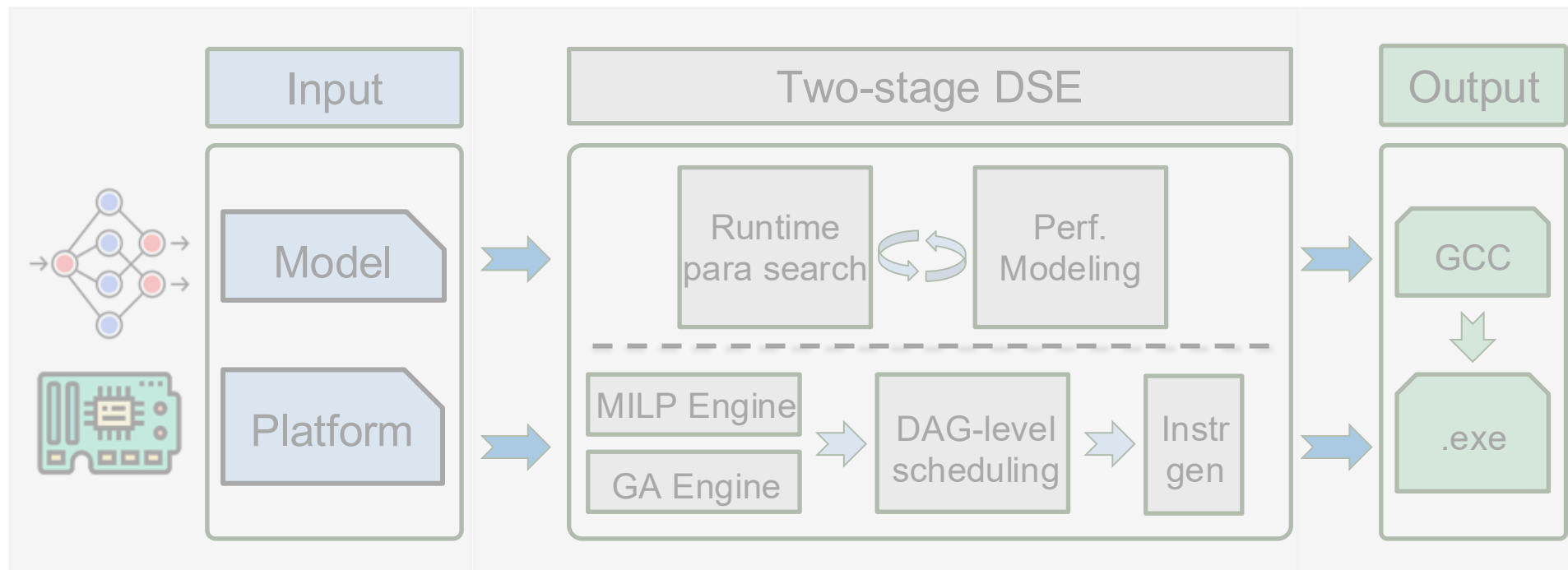
DORA Framework

Input:

- 1) MM-based AI Model;
- 2) Hardware platforms information.

Output:

- 1) Bitstream running on the AIE and PL;
- 2) Executable file running on ARM CPU.



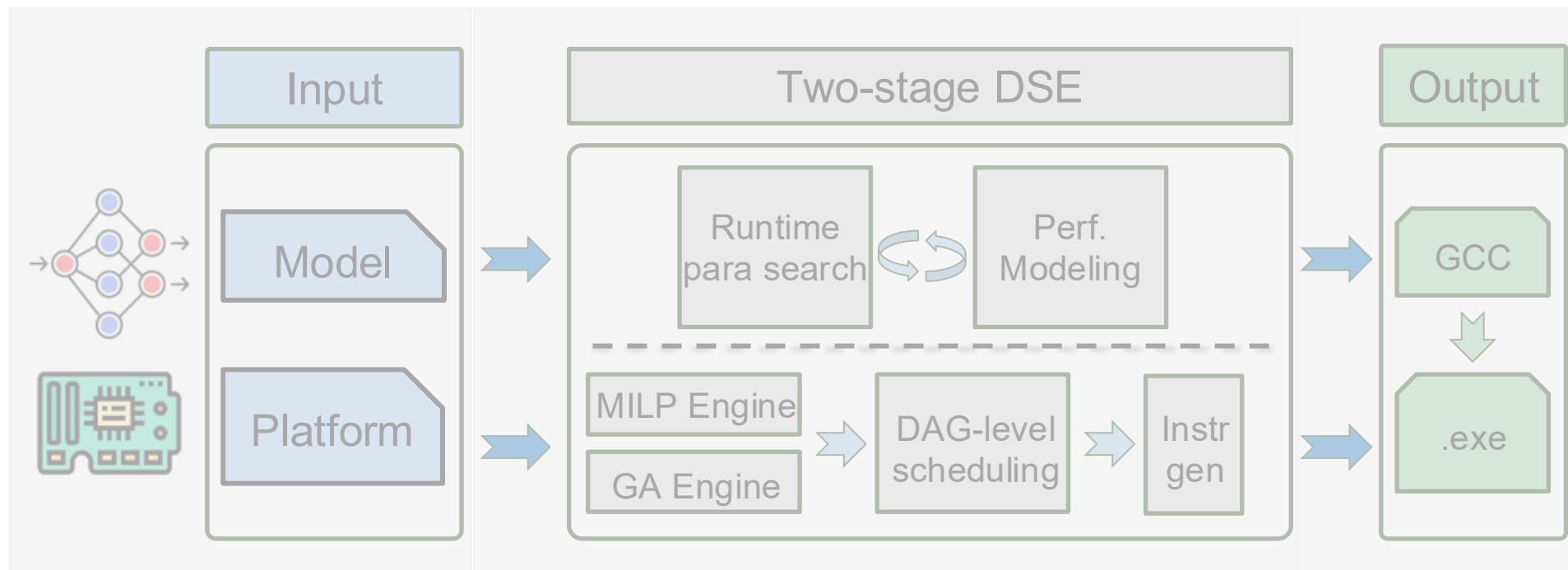
DORA Framework

Input:

- 1) MM-based AI Model;
- 2) Hardware platforms information.

Output:

- 1) Bitstream running on the AIE and PL;
- 2) Executable file running on ARM CPU.



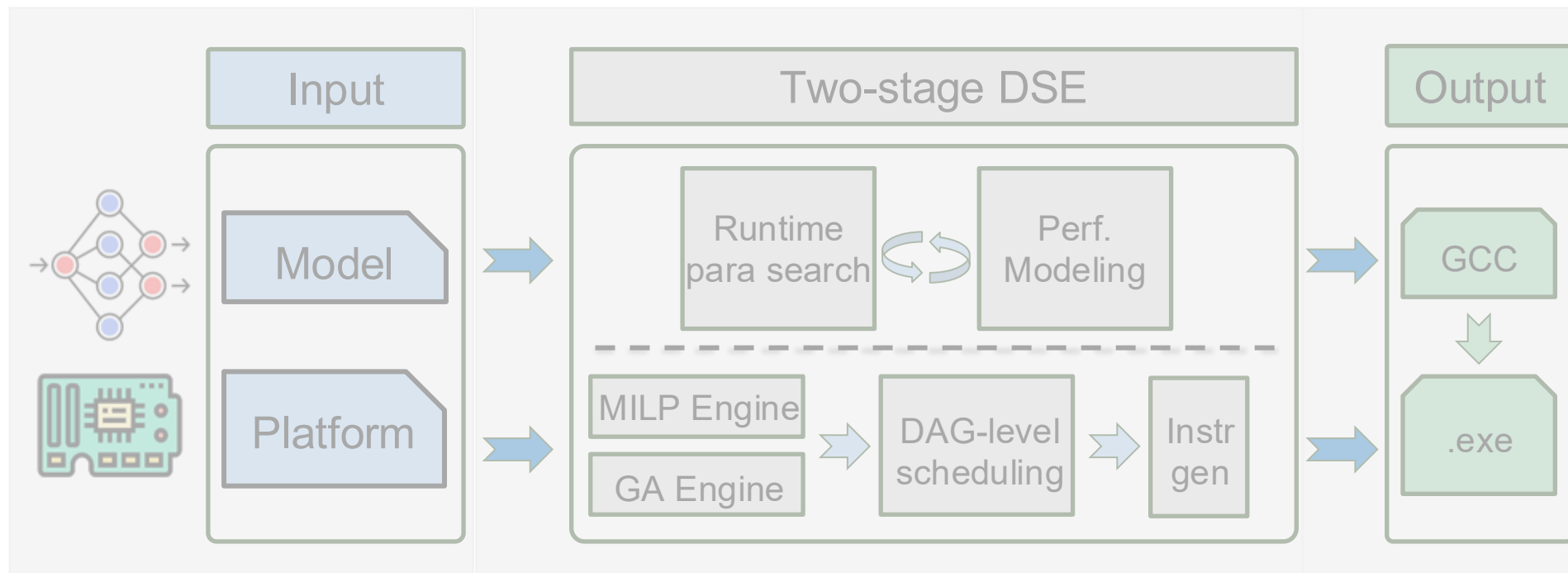
DORA Framework

Input:

- 1) MM-based AI Model;
- 2) Hardware platforms information.

Output:

- 1) Bitstream running on the AIE and PL;
- 2) Executable file running on ARM CPU.



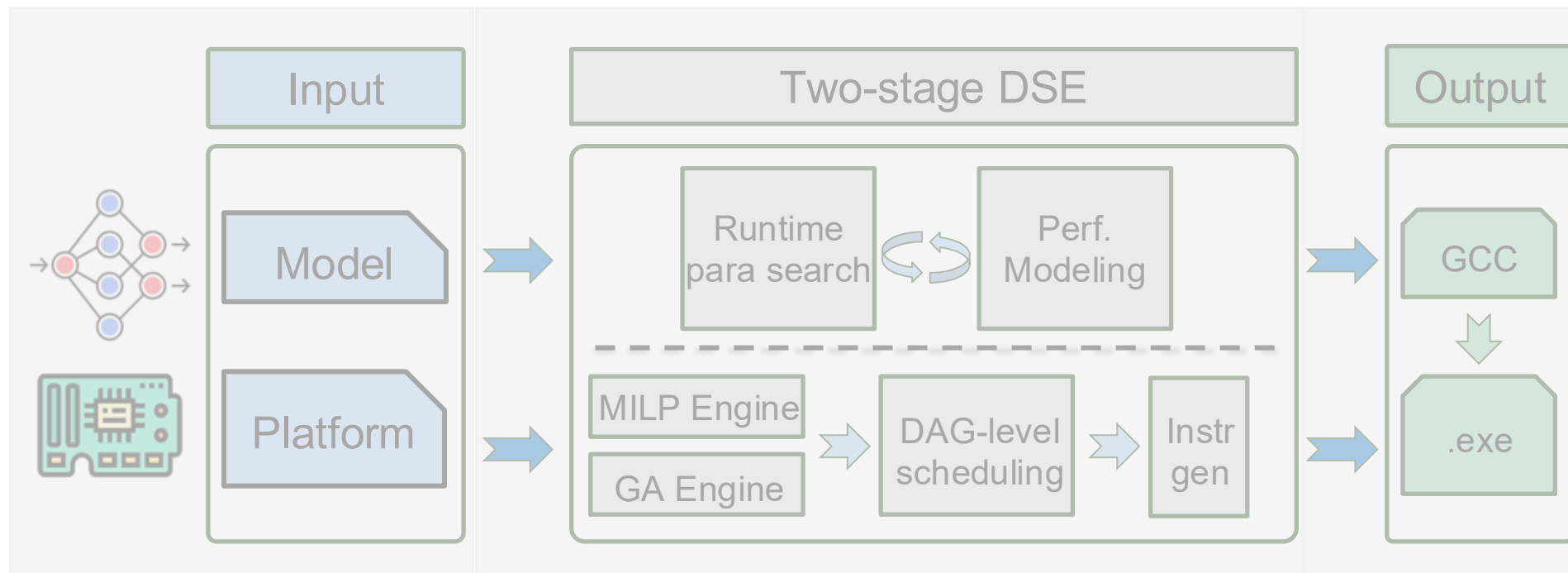
DORA Framework

Input:

- 1) MM-based AI Model;
- 2) Hardware platforms information.

Output:

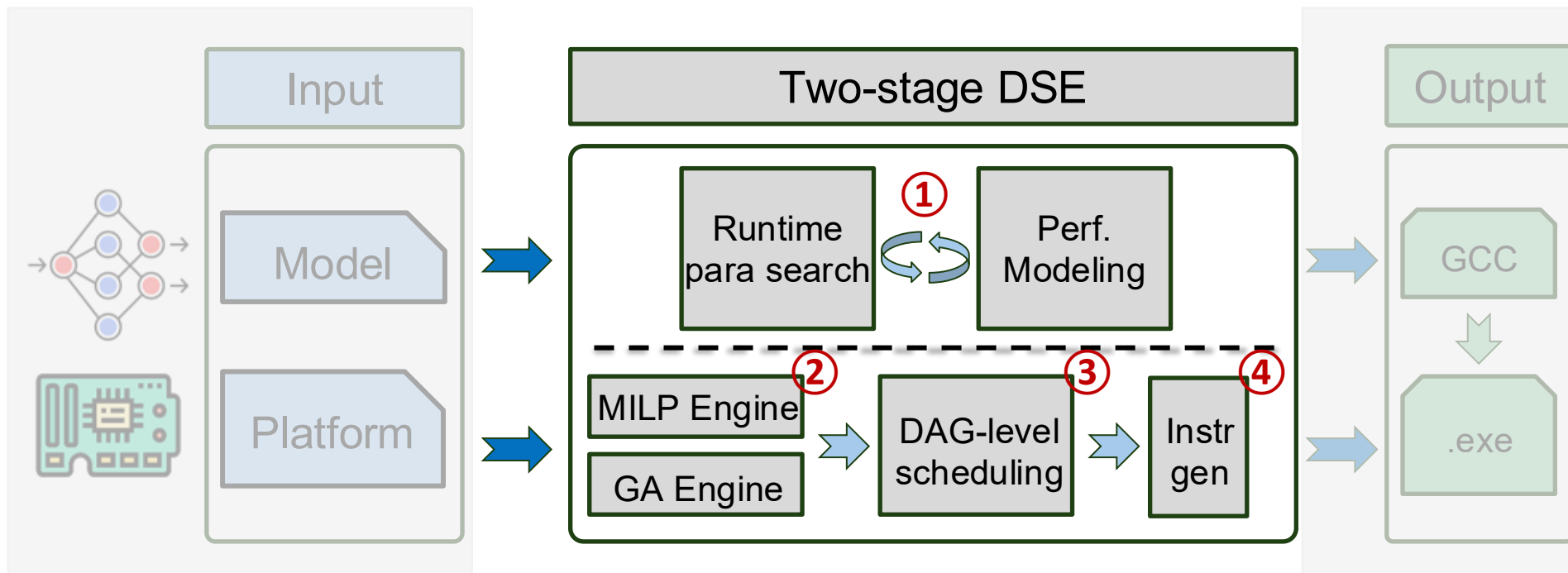
- 1) Bitstream running on the AIE and PL;
- 2) Executable file running on ARM CPU.



DORA Framework

① Single-layer brute-force design space exploration.

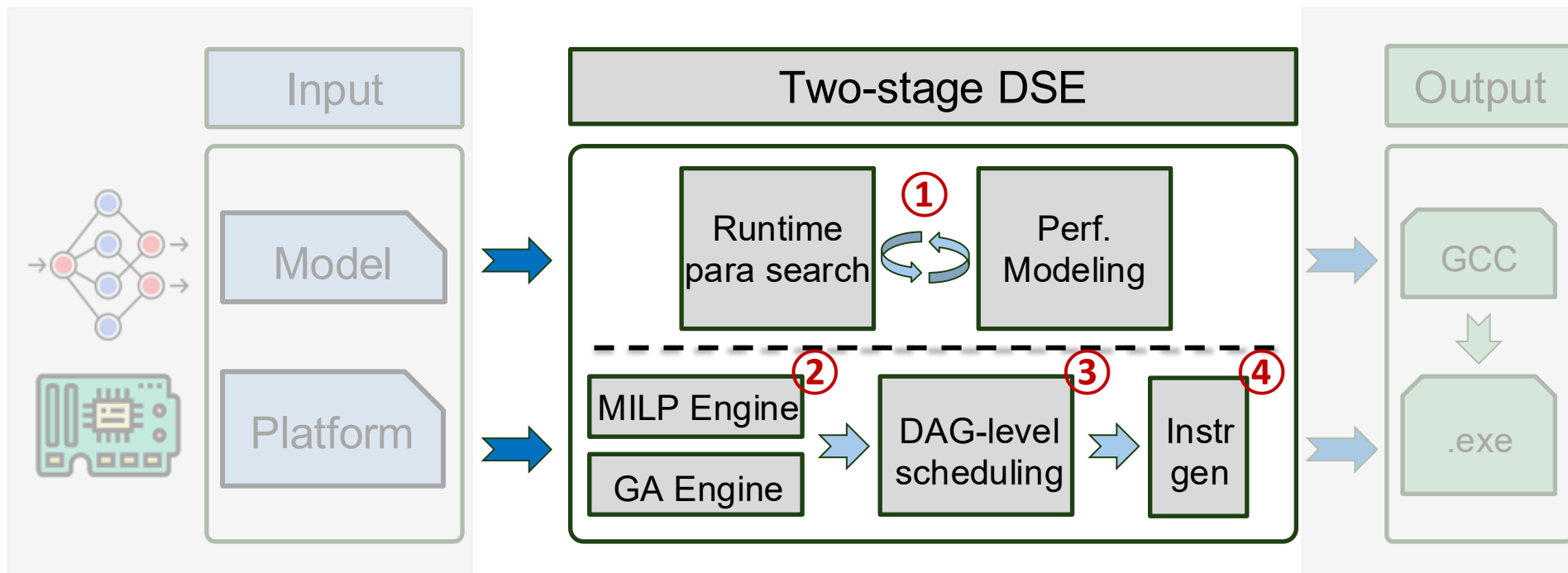
- Populate all possible mapping solutions with #memory and #compute resources for a single layer.
- Store the candidate execution table for the second stage.



DORA Framework

① Single-layer brute-force design space exploration.

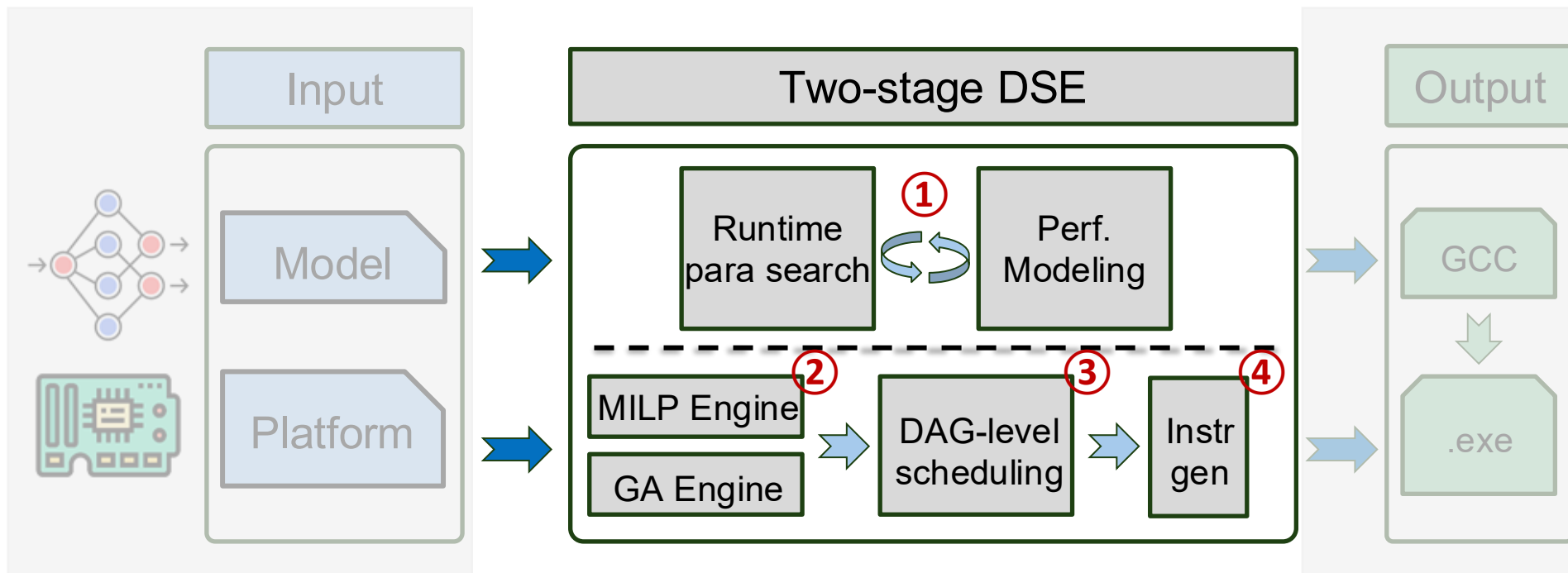
- Populate all possible mapping solutions with #memory and #compute resources for a single layer.
- Store the candidate execution table for the second stage.



DORA Framework

① Single-layer brute-force design space exploration.

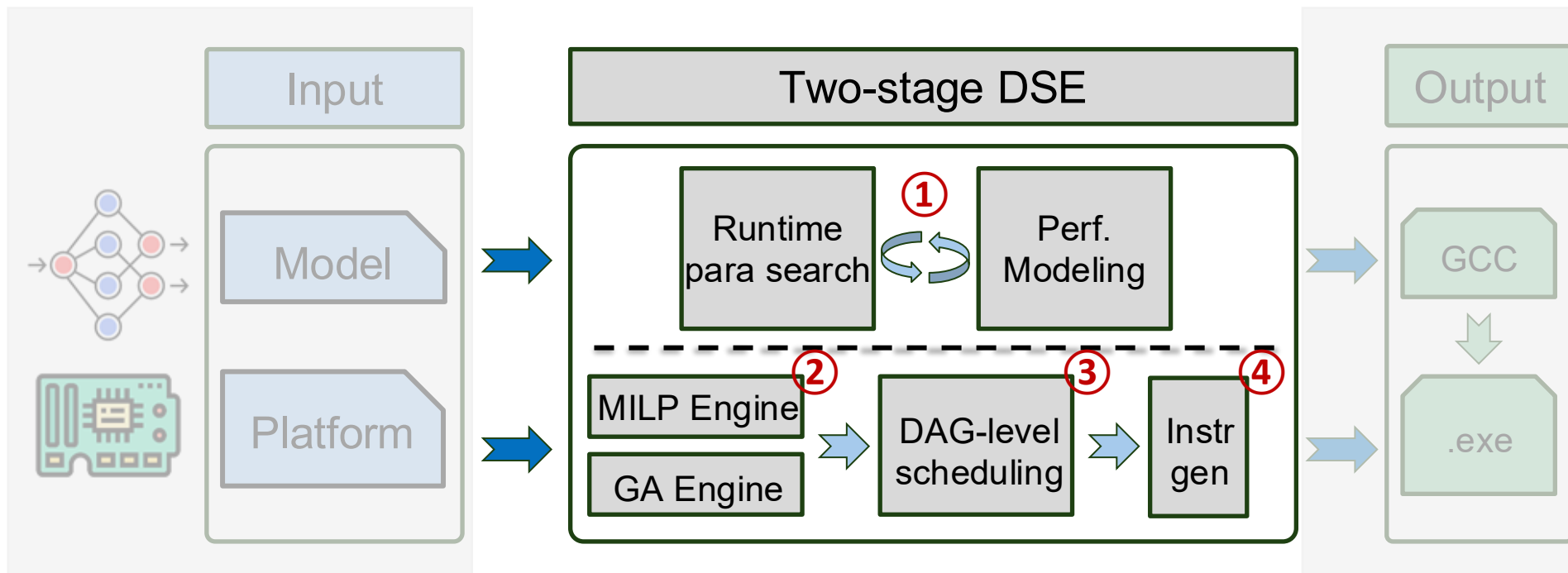
- Populate all possible mapping solutions with #memory and #compute resources for a single layer.
- Store the candidate execution table for the second stage.



DORA Framework

① Single-layer brute-force design space exploration.

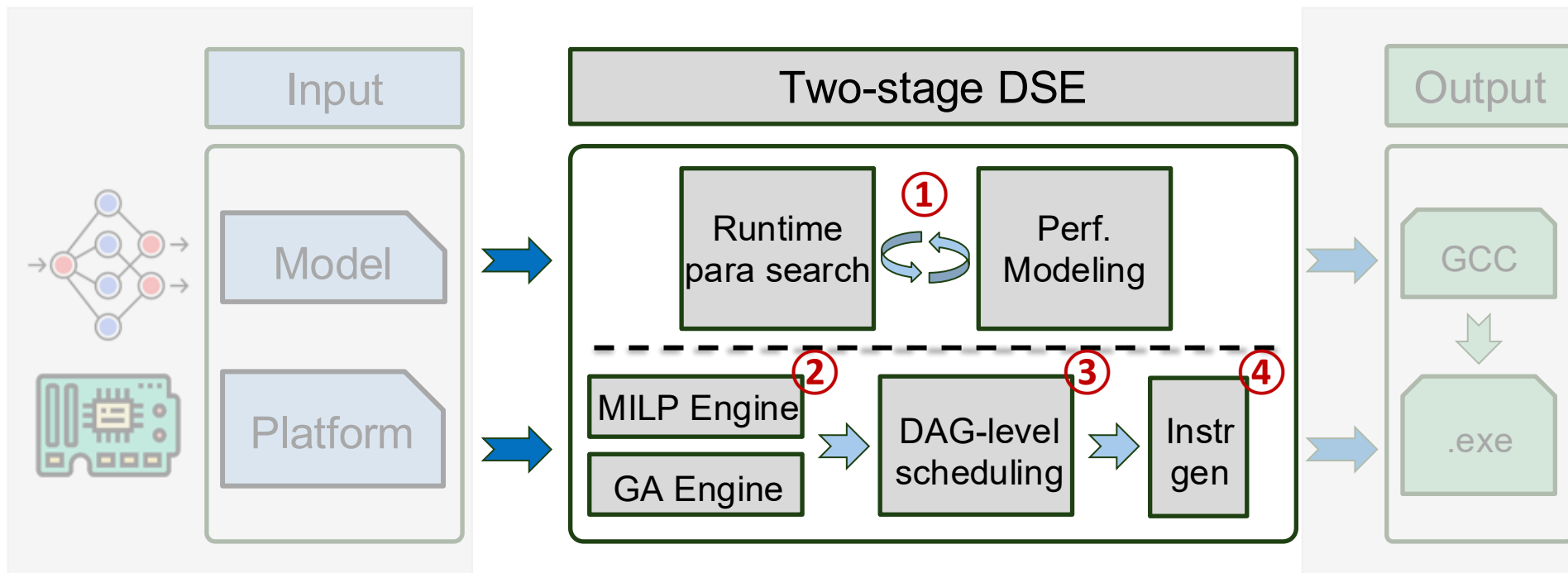
- Populate all possible mapping solutions with #memory and #compute resources for a single layer.
- Store the candidate execution table for the second stage.



DORA Framework

① Single-layer brute-force design space exploration.

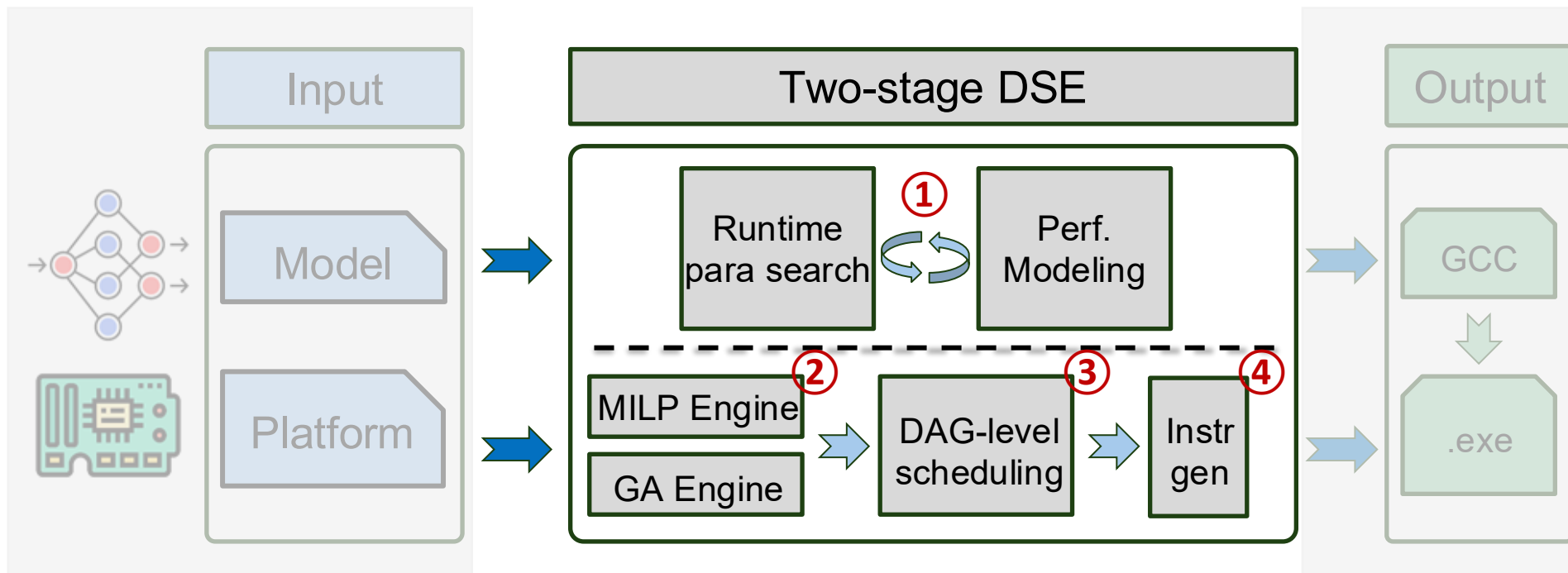
- Populate all possible mapping solutions with #memory and #compute resources for a single layer.
- Store the candidate execution table for the second stage.



DORA Framework

② Scheduling engines at the graph level for whole workloads.

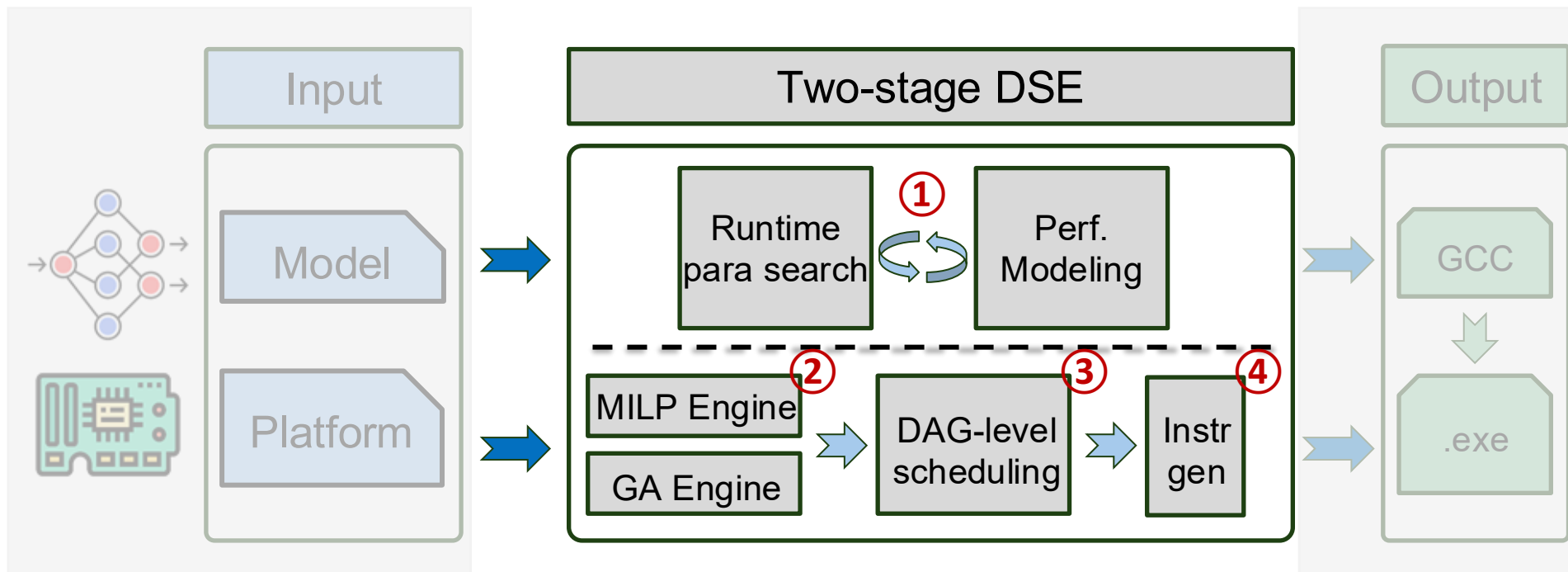
- Mixed Integer Linear Programming (MILP) engine can guarantee the optimal scheduling.
- Genetic Algorithm (GA) engine can achieve high efficiency.



DORA Framework

② Scheduling engines at the graph level for whole workloads.

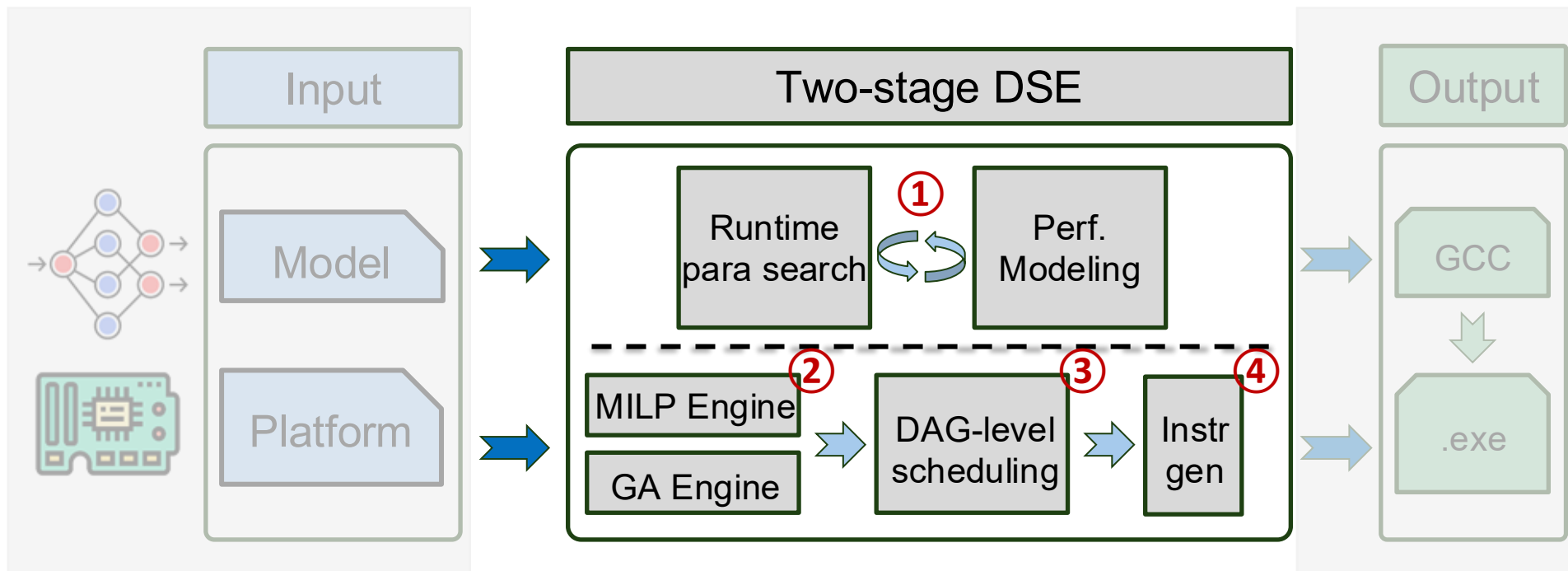
- Mixed Integer Linear Programming (MILP) engine can guarantee the optimal scheduling.
- Genetic Algorithm (GA) engine can achieve high efficiency.



DORA Framework

② Scheduling engines at the graph level for whole workloads.

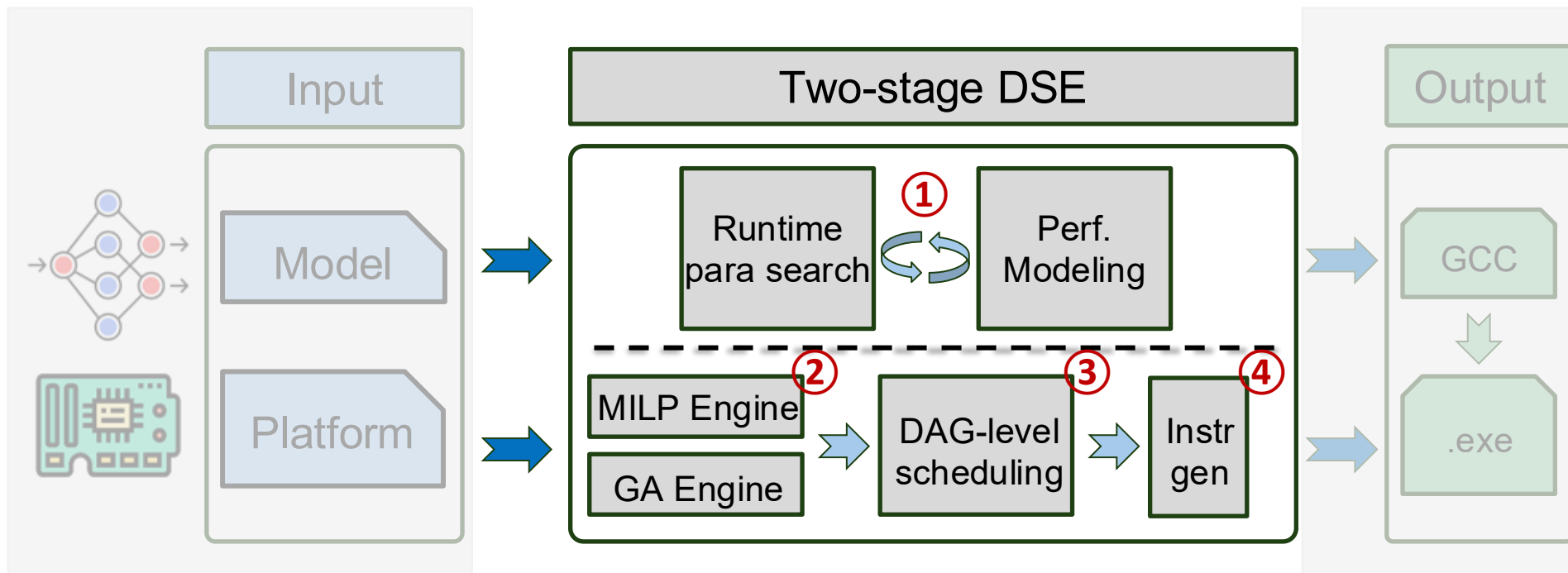
- Mixed Integer Linear Programming (MILP) engine can guarantee the optimal scheduling.
- Genetic Algorithm (GA) engine can achieve high efficiency.



DORA Framework

③ Generate execution flow for each functional unit.

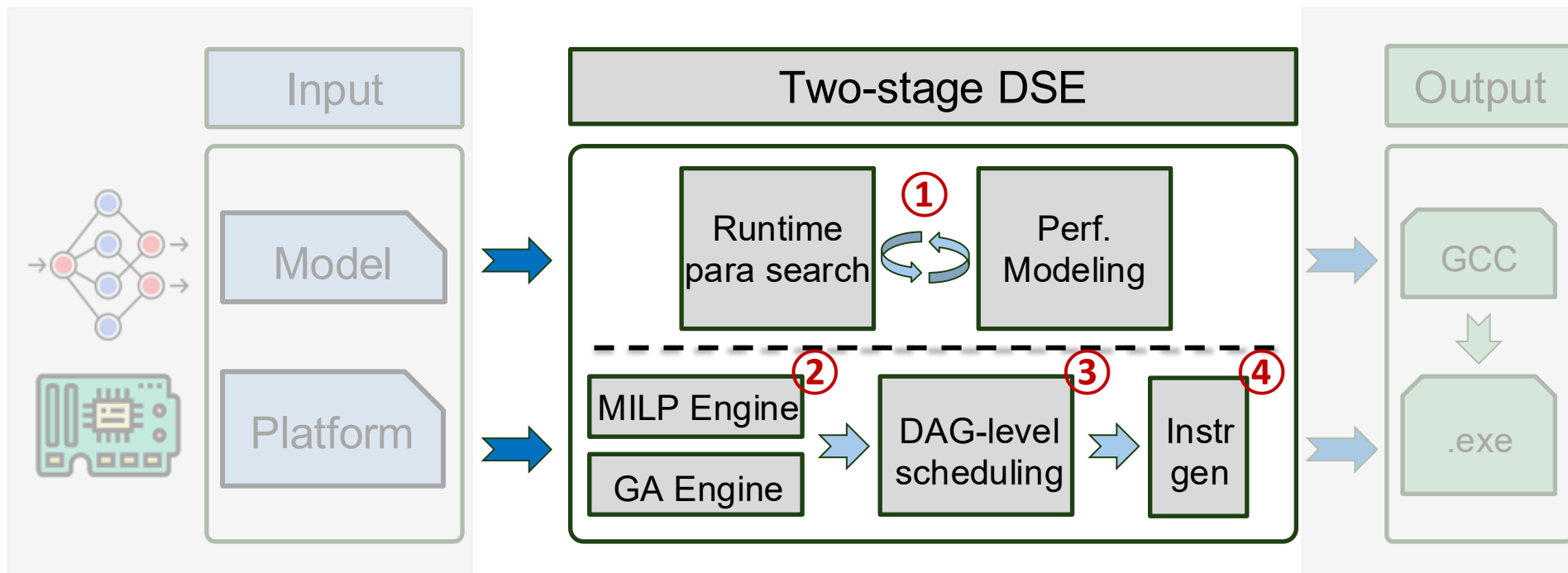
- Analyze the dataflow for each functional unit;
- Prepare instruction sequence at FU level.



DORA Framework

③ Generate execution flow for each functional unit.

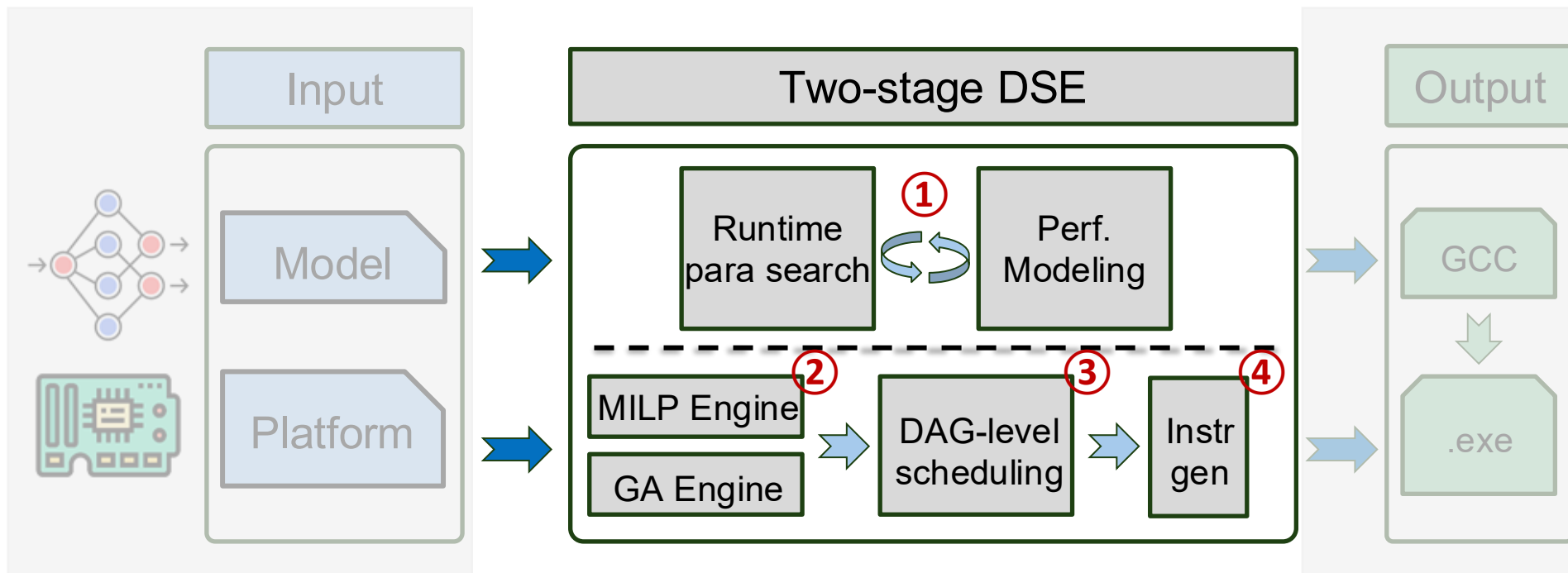
- Analyze the dataflow for each functional unit;
- Prepare instruction sequence at FU level.



DORA Framework

③ Generate execution flow for each functional unit.

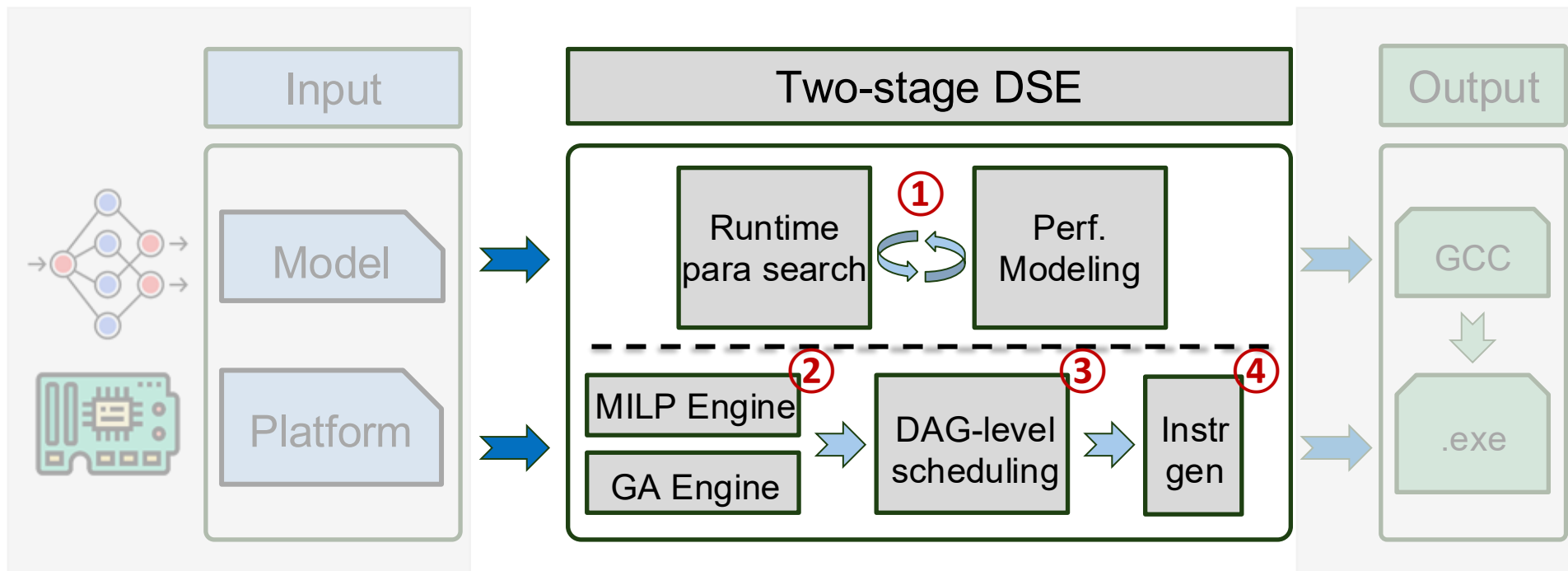
- Analyze the dataflow for each functional unit;
- Prepare instruction sequence at FU level.



DORA Framework

④ Instruction sequence generation.

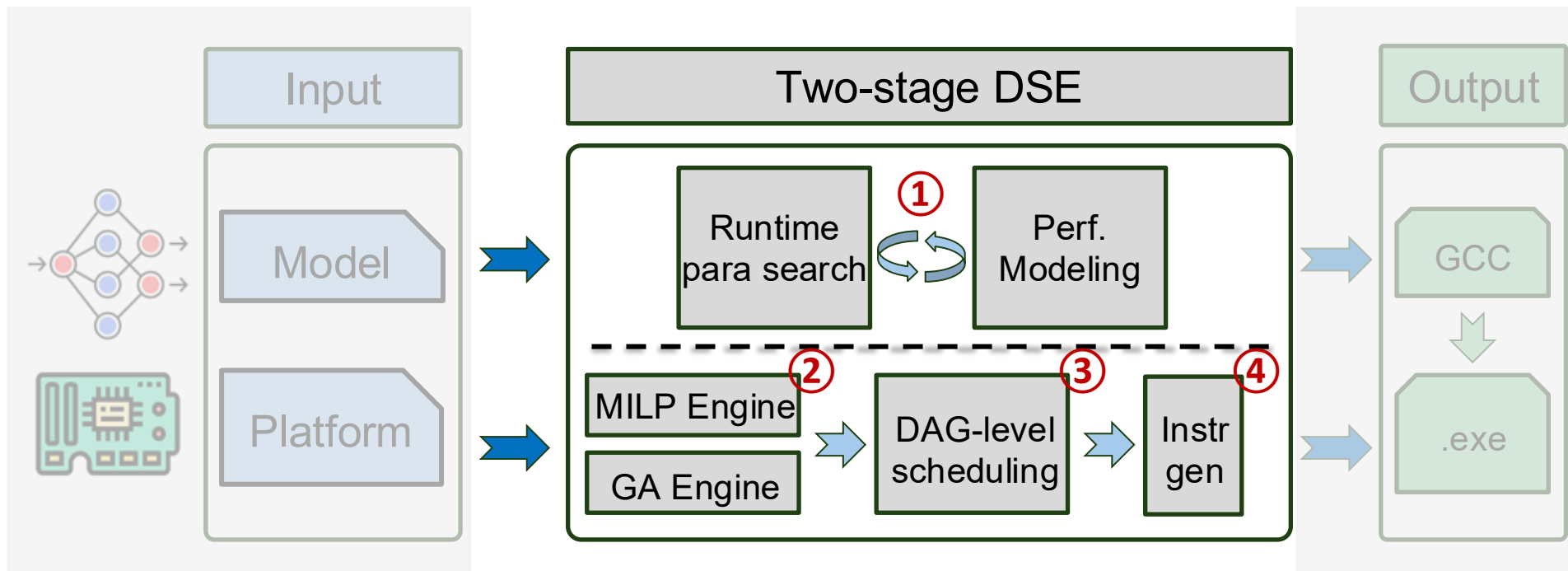
- Generate the instruction sequence and store them into host DRAM.



DORA Framework

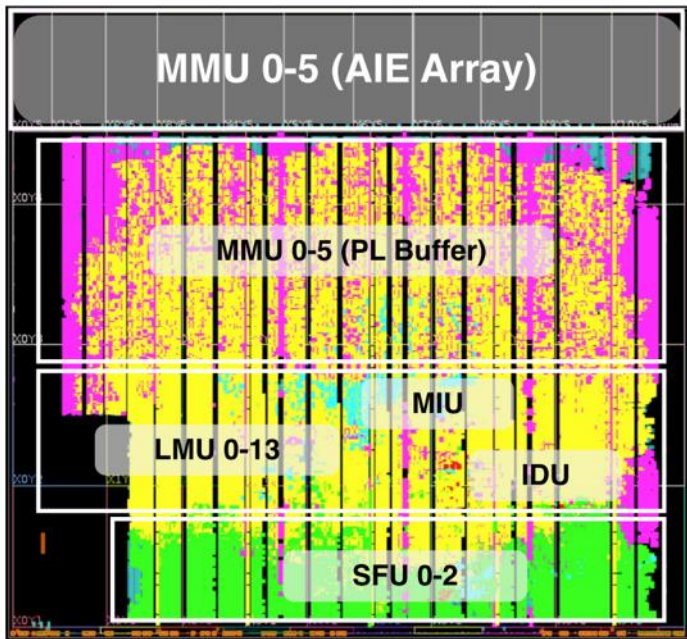
④ Instruction sequence generation.

- Generate the instruction sequence and store them into host DRAM.



Experiment Setup

- We prototype DORA architecture on AMD Versal VCK190 platform.
- Instruction-related modules only consume REG and LUT, which are usually not bottlenecks in accelerator design.



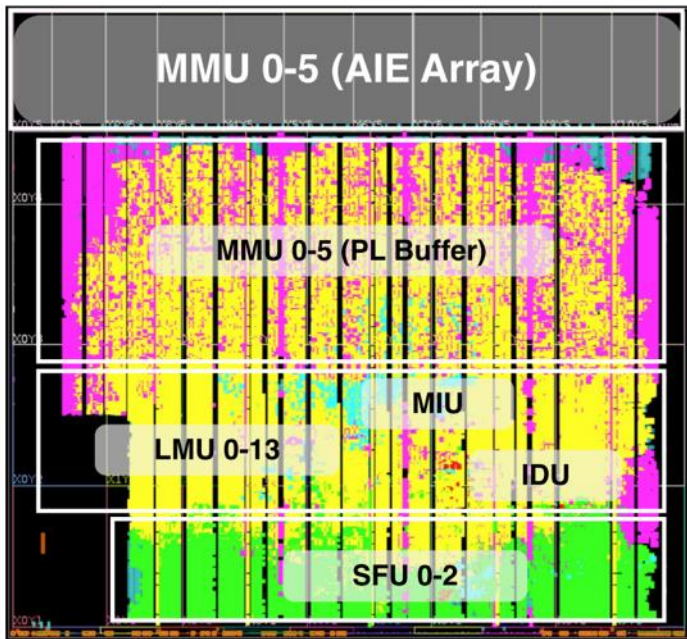
DORA implementation layout

Unit	REG	LUTLogic	LUTMem	BRAM	URAM	DSP	AIE
MMU	110598 (6.12%)	103890 (11.52%)	0	576 (60%)	0	6 (0.3%)	384 (96%)
LMU	64358 (3.64%)	174538 (19.46%)	43358 (9.66%)	0	448 (96%)	280 (14.28%)	0
SFU	72000 (4%)	100974 (11.22%)	2328 (0.5%)	48 (4.95%)	0	96 (4.89%)	0
IDU	613 (0.03%)	415 (0.05%)	0	0	0	0	0
MIU	1810 (0.09%)	3913 (0.44%)	28 (<0.1%)	0	0	6 (0.3%)	0
Fully-con Network	264768 (14.7%)	68208 (7.58%)	0	0	0	0	0
Others	55111 (3%)	65504 (7.28%)	1419 (0.3%)	23 (2.4%)	0	0	0
Total	569258 (31.5%)	517442 (57.4%)	47133 (10.5%)	647 (67.4%)	448 (96%)	388 (19.8%)	384 (96%)

Resource utilization on VCK190

Experiment Setup

- We prototype DORA architecture on AMD Versal VCK190 platform.
- Instruction-related modules only consume REG and LUT, which are usually not bottlenecks in accelerator design.



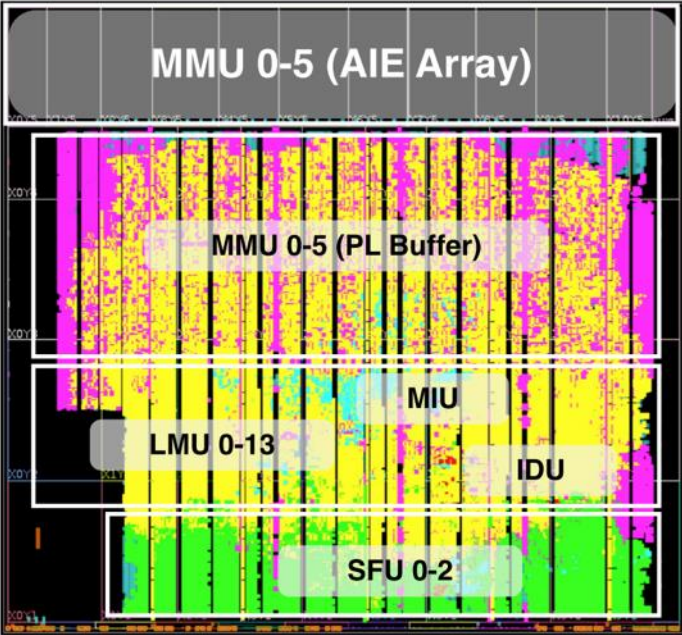
DORA implementation layout

Unit	REG	LUTLogic	LUTMem	BRAM	URAM	DSP	AIE
MMU	110598 (6.12%)	103890 (11.52%)	0	576 (60%)	0	6 (0.3%)	384 (96%)
LMU	64358 (3.64%)	174538 (19.46%)	43358 (9.66%)	0	448 (96%)	280 (14.28%)	0
SFU	72000 (4%)	100974 (11.22%)	2328 (0.5%)	48 (4.95%)	0	96 (4.89%)	0
IDU	613 (0.03%)	415 (0.05%)	0	0	0	0	0
MIU	1810 (0.09%)	3913 (0.44%)	28 (<0.1%)	0	0	6 (0.3%)	0
Fully-con Network	264768 (14.7%)	68208 (7.58%)	0	0	0	0	0
Others	55111 (3%)	65504 (7.28%)	1419 (0.3%)	23 (2.4%)	0	0	0
Total	569258 (31.5%)	517442 (57.4%)	47133 (10.5%)	647 (67.4%)	448 (96%)	388 (19.8%)	384 (96%)

Resource utilization on VCK190

Experiment Setup

- We prototype DORA architecture on AMD Versal VCK190 platform.
- Instruction-related modules only consume REG and LUT, which are usually not bottlenecks in accelerator design.



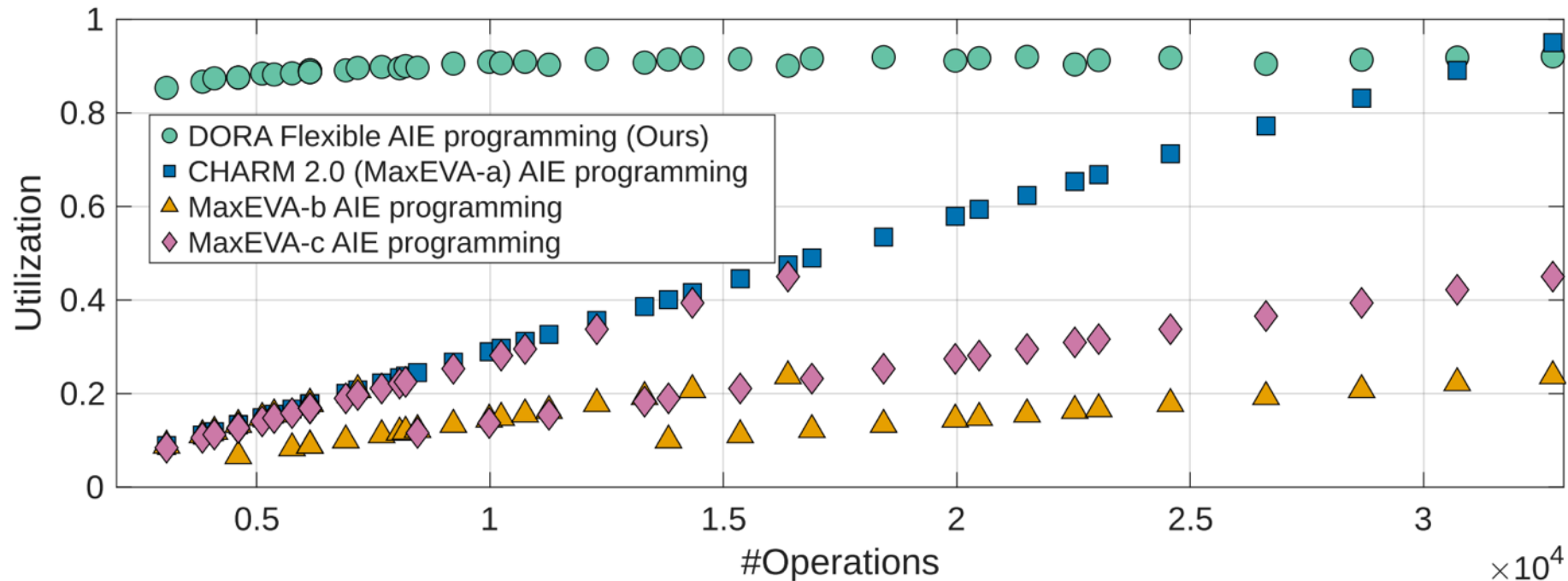
DORA implementation layout

Unit	REG	LUTLogic	LUTMem	BRAM	URAM	DSP	AIE
MMU	110598 (6.12%)	103890 (11.52%)	0	576 (60%)	0	6 (0.3%)	384 (96%)
LMU	64358 (3.64%)	174538 (19.46%)	43358 (9.66%)	0	448 (96%)	280 (14.28%)	0
SFU	72000 (4%)	100974 (11.22%)	2328 (0.5%)	48 (4.95%)	0	96 (4.89%)	0
IDU	613 (0.03%)	415 (0.05%)	0	0	0	0	0
MIU	1810 (0.09%)	3913 (0.44%)	28 (<0.1%)	0	0	6 (0.3%)	0
Fully-con Network	264768 (14.7%)	68208 (7.58%)	0	0	0	0	0
Others	55111 (3%)	65504 (7.28%)	1419 (0.3%)	23 (2.4%)	0	0	0
Total	569258 (31.5%)	517442 (57.4%)	47133 (10.5%)	647 (67.4%)	448 (96%)	388 (19.8%)	384 (96%)

Resource utilization on VCK190

Experiment Results

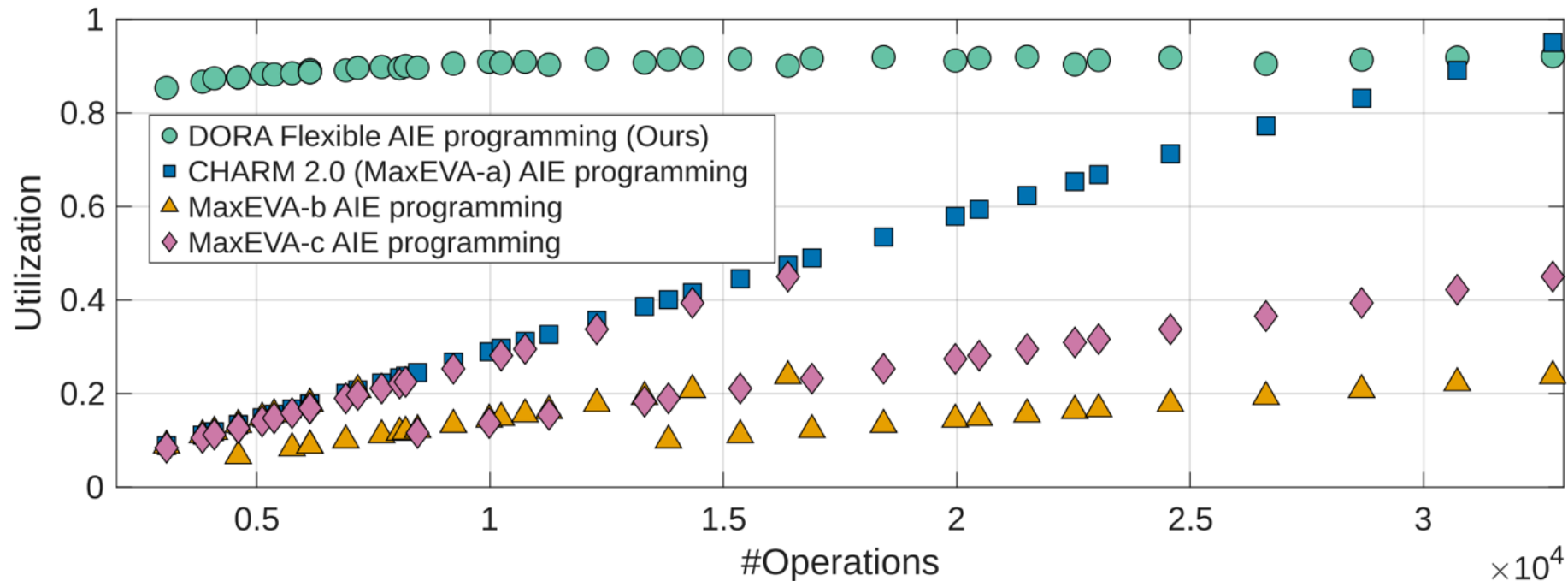
- Flexibility on a single AI Engine core for diverse matrix multiplication.
- Populate MM operations with different sizes, and DORA parallelism management efficiently handles diverse MMs.



Experiments on a single AIE tile for diverse matrix multiplications.

Experiment Results

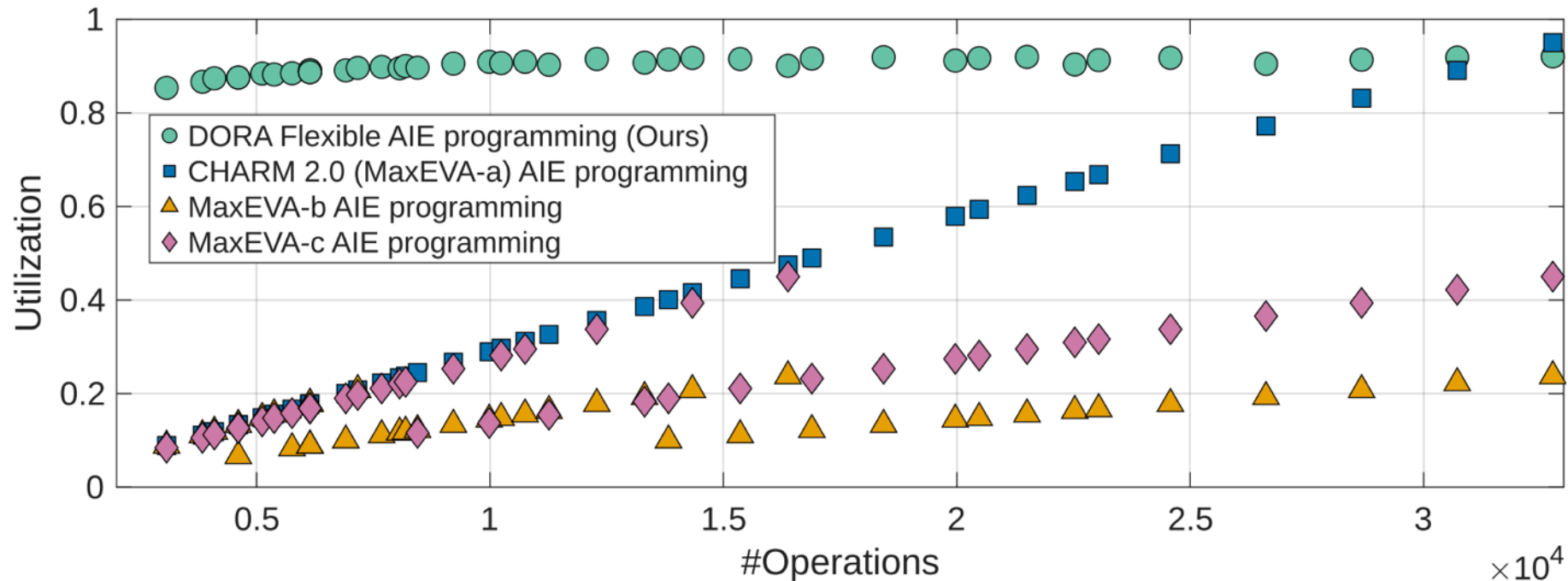
- Flexibility on a single AI Engine core for diverse matrix multiplication.
- Populate MM operations with different sizes, and DORA parallelism management efficiently handles diverse MMs.



Experiments on a single AIE tile for diverse matrix multiplications.

Experiment Results

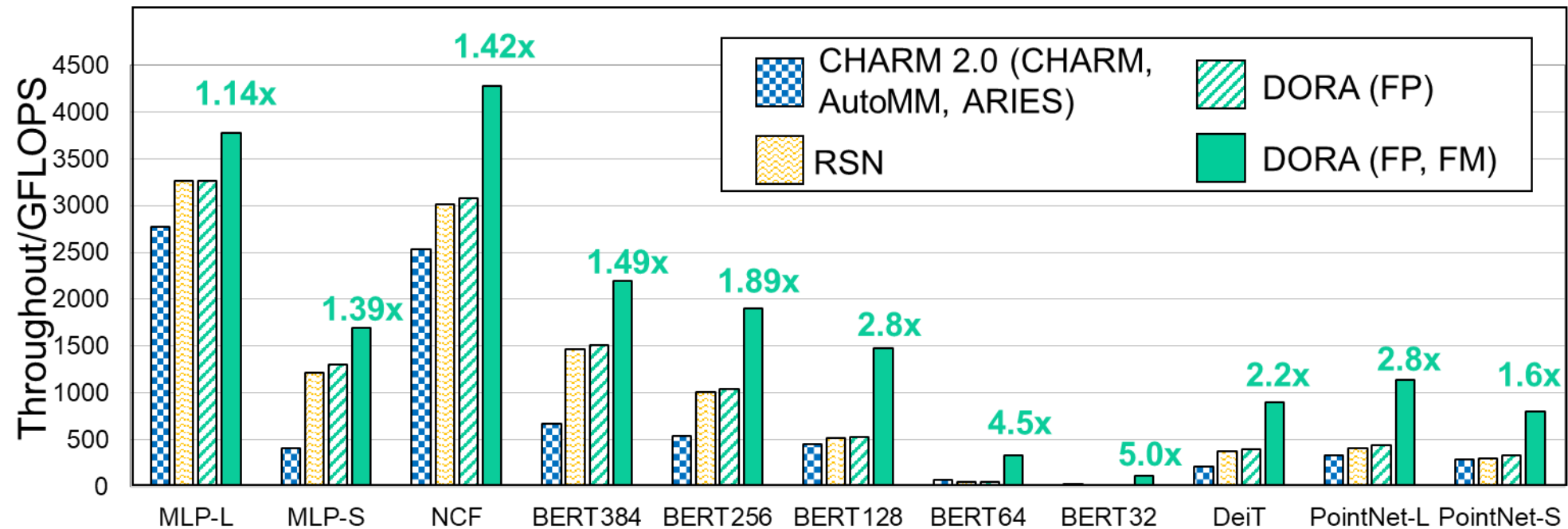
- Flexibility on a single AI Engine core for diverse matrix multiplication.
- Populate MM operations with different sizes, and DORA parallelism management efficiently handles diverse MMs.



Experiments on a single AIE tile for diverse matrix multiplications.

Experiment Results

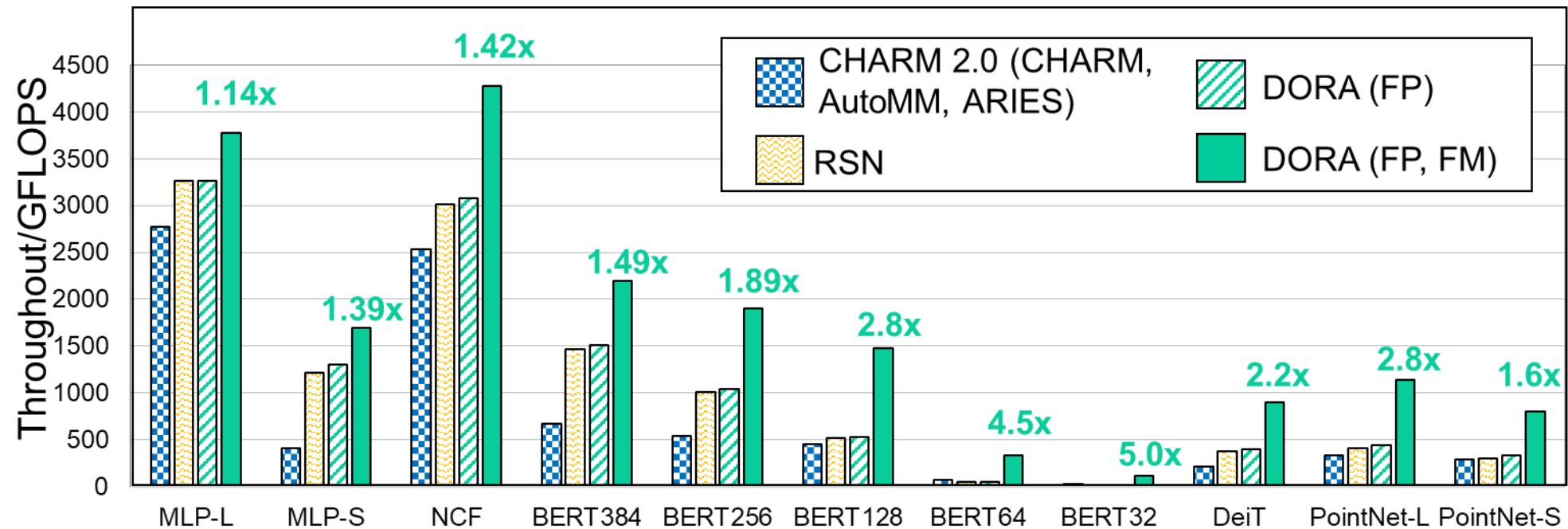
- Efficiency on end-to-end DNN inference.
- DORA always outperforms customized architecture design and existing overlay design.



Experiments on end-to-end DNN inference for diverse models.

Experiment Results

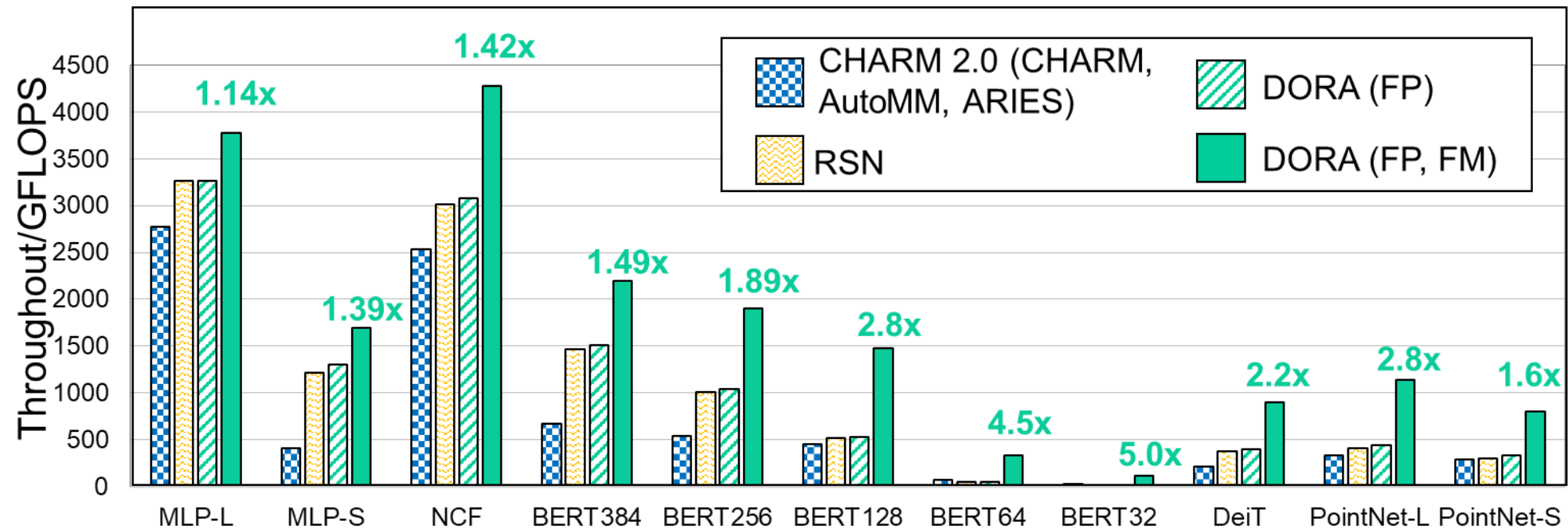
- Efficiency on end-to-end DNN inference.
- DORA always outperforms customized architecture design and existing overlay design.



Experiments on end-to-end DNN inference for diverse models.

Experiment Results

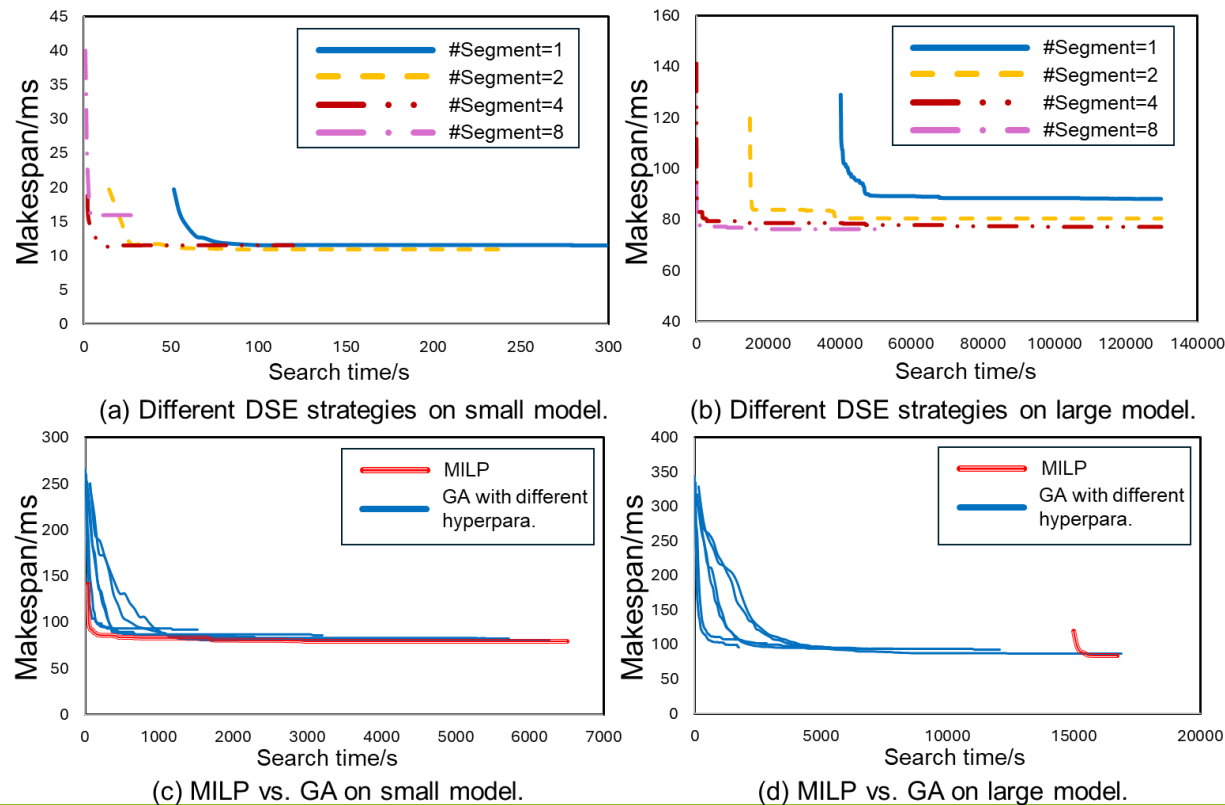
- Efficiency on end-to-end DNN inference.
- DORA always outperforms customized architecture design and existing overlay design.



Experiments on end-to-end DNN inference for diverse models.

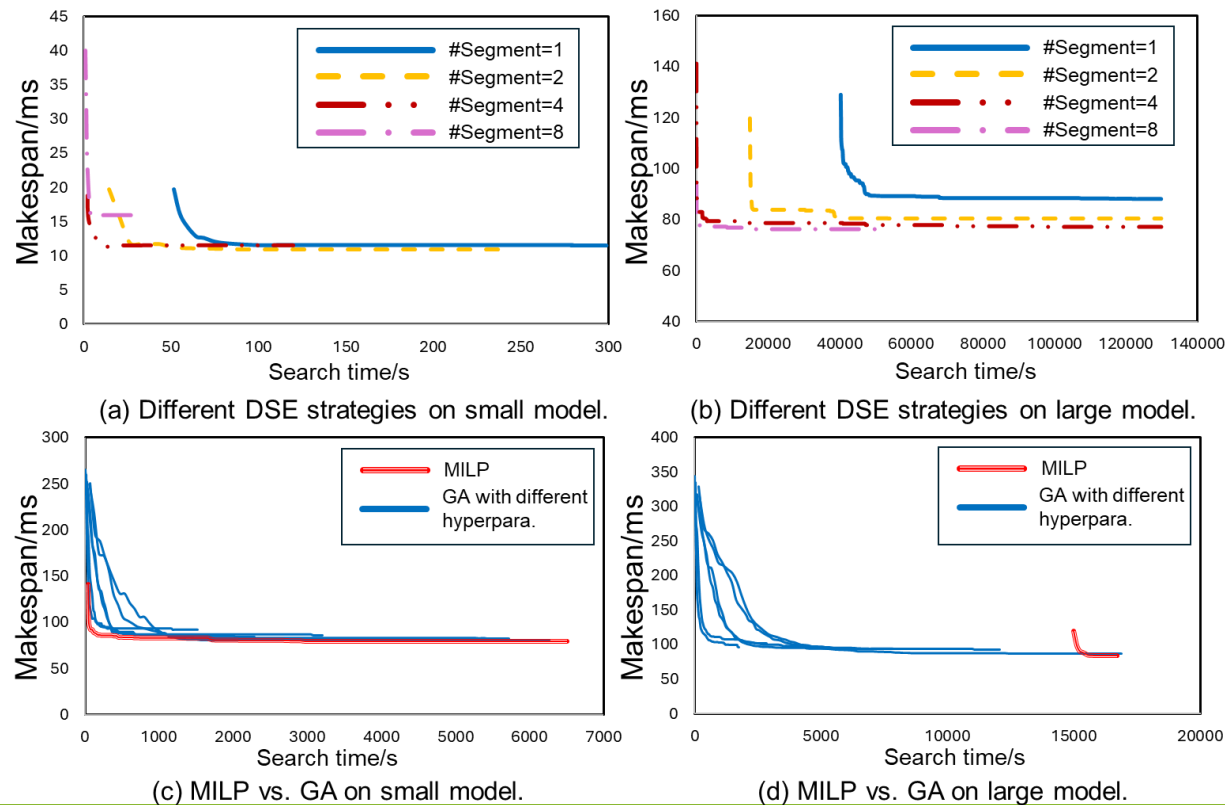
Experiment Results

- Optimality and efficiency on MILP and GA formulations.
 - MILP guarantees the **optimality** of medium problem size.
 - GA provides the **scalability** and **efficiency** for large problem size.



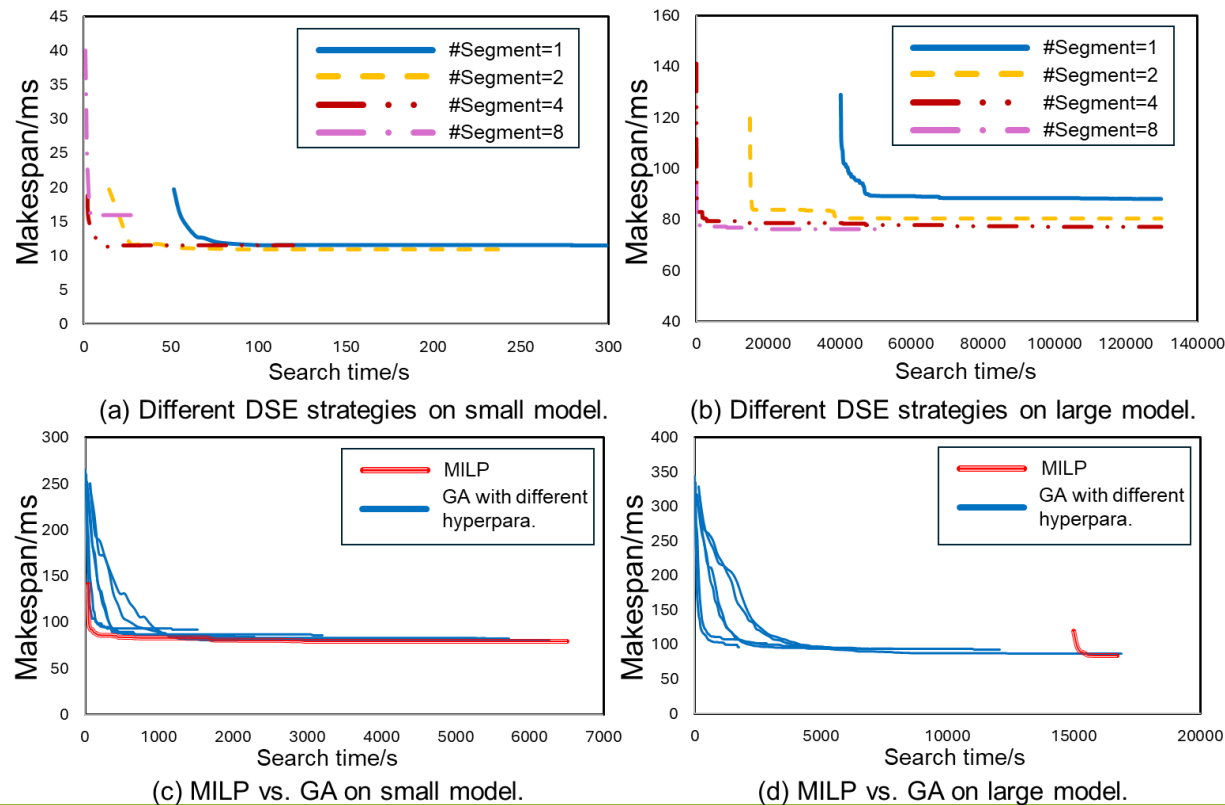
Experiment Results

- Optimality and efficiency on MILP and GA formulations.
 - MILP guarantees the **optimality** of medium problem size.
 - GA provides the **scalability** and **efficiency** for large problem size.



Experiment Results

- Optimality and efficiency on MILP and GA formulations.
 - MILP guarantees the **optimality** of medium problem size.
 - GA provides the **scalability** and **efficiency** for large problem size.



Key Takeaways

- DORA proposes a **flexible overlay architecture** that orchestrates dataflow with instructions for **diverse workloads**.
- DORA features an **automatic framework** with 2-stage DSE that can take applications and platform information to generate the **ready-to-run executable files and bitstreams**.
- DORA has a good **generality** and **scalability**, especially on large problem sizes.
- DORA open-source GitHub repo:
<https://github.com/arc-research-lab/DORA.git>

Key Takeaways

- DORA proposes a **flexible overlay architecture** that orchestrates dataflow with instructions for **diverse workloads**.
- DORA features an **automatic framework** with 2-stage DSE that can take applications and platform information to generate the **ready-to-run executable files and bitstreams**.
- DORA has a good **generality** and **scalability**, especially on large problem sizes.
- DORA open-source GitHub repo:
<https://github.com/arc-research-lab/DORA.git>

Thank You & Welcome to questions!

DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Xingzhen Chen, Zhuoping Yang, Jinming Zhuang, Shixin Ji, Sarah Schultz,
Zheng Dong*, Weisong Shi[†] and Peipei Zhou

Brown University; Wayne State University*; University of Delaware[†]

xingzhen_chen@brown.edu

peipei_zhou@brown.edu

<https://peipeizhou-eecs.github.io/>

<https://github.com/arc-research-lab/DORA.git>



DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration

Xingzhen Chen, Zhuoping Yang, Jinming Zhuang, Shixin Ji, Sarah Schultz,
Zheng Dong*, Weisong Shi[†] and Peipei Zhou

Brown University; Wayne State University*; University of Delaware[†]

xingzhen_chen@brown.edu

peipei_zhou@brown.edu

<https://peipeizhou-eecs.github.io/>

<https://github.com/arc-research-lab/DORA.git>



U.S. DEPARTMENT
of ENERGY



BROWN



WAYNE STATE
UNIVERSITY

UNIVERSITY OF
DELAWARE